MILITARY SPECIFICATION SHEET

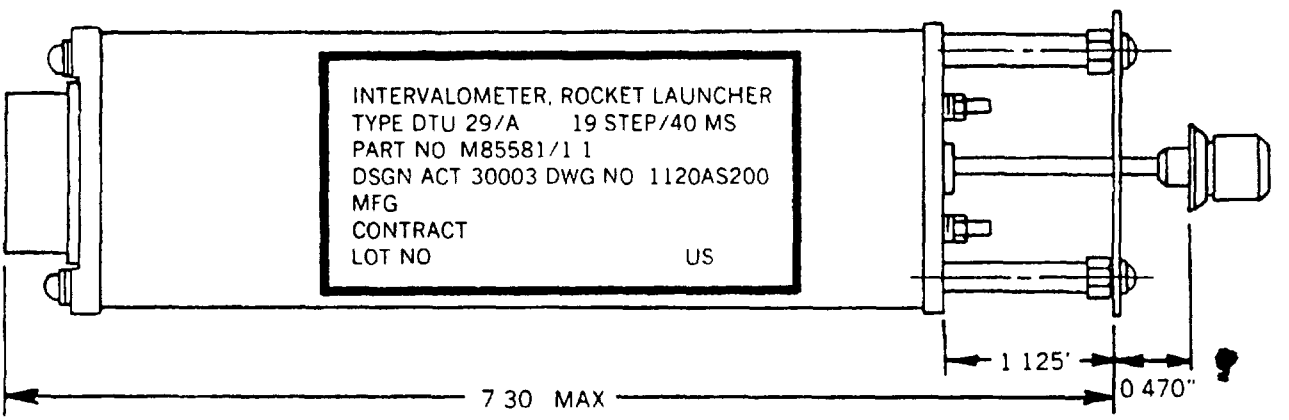
# INTERVALOMETER, ROCKET LAUNCHER, REUSABLE, NINETEEN (19) OUTPUT, DIRECT LINE OPERATED

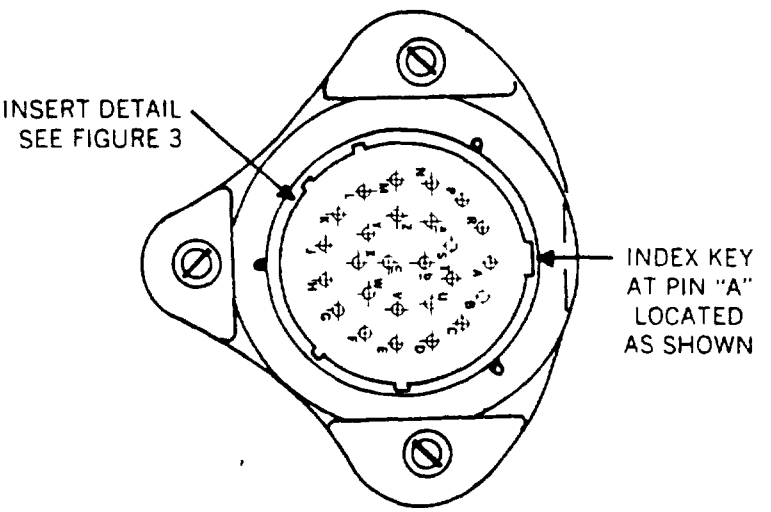
This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the intervalometers described herein shall consist of this document and the latest issue of MIL-I-85581.

This specification sheet covers the following equipment:

| Type number | Part number |
|---|---|
| DTU-29/A | M85581/1-1 |




NOTES

1 KNOB ROTATION CCW WHEN VIEWED FROM KNOB END
2 ALL INTERVALOMETERS SHALL BE CHECKED ELECTRICALLY TO INSURE INDEX POSITION OF KNOB IS ALIGNED WITH "LOAD" POSITION OF LOAD/ARM PLATE (NAVAIRSYSCOM DRAWING NO 1120AS246)

FIGURE 1. Intervalometer.
(For exact dimensions refer to NAVAIRSYSCOM Drawing No. 1120AS200, Intervalometer Assembly)



FSC 5930

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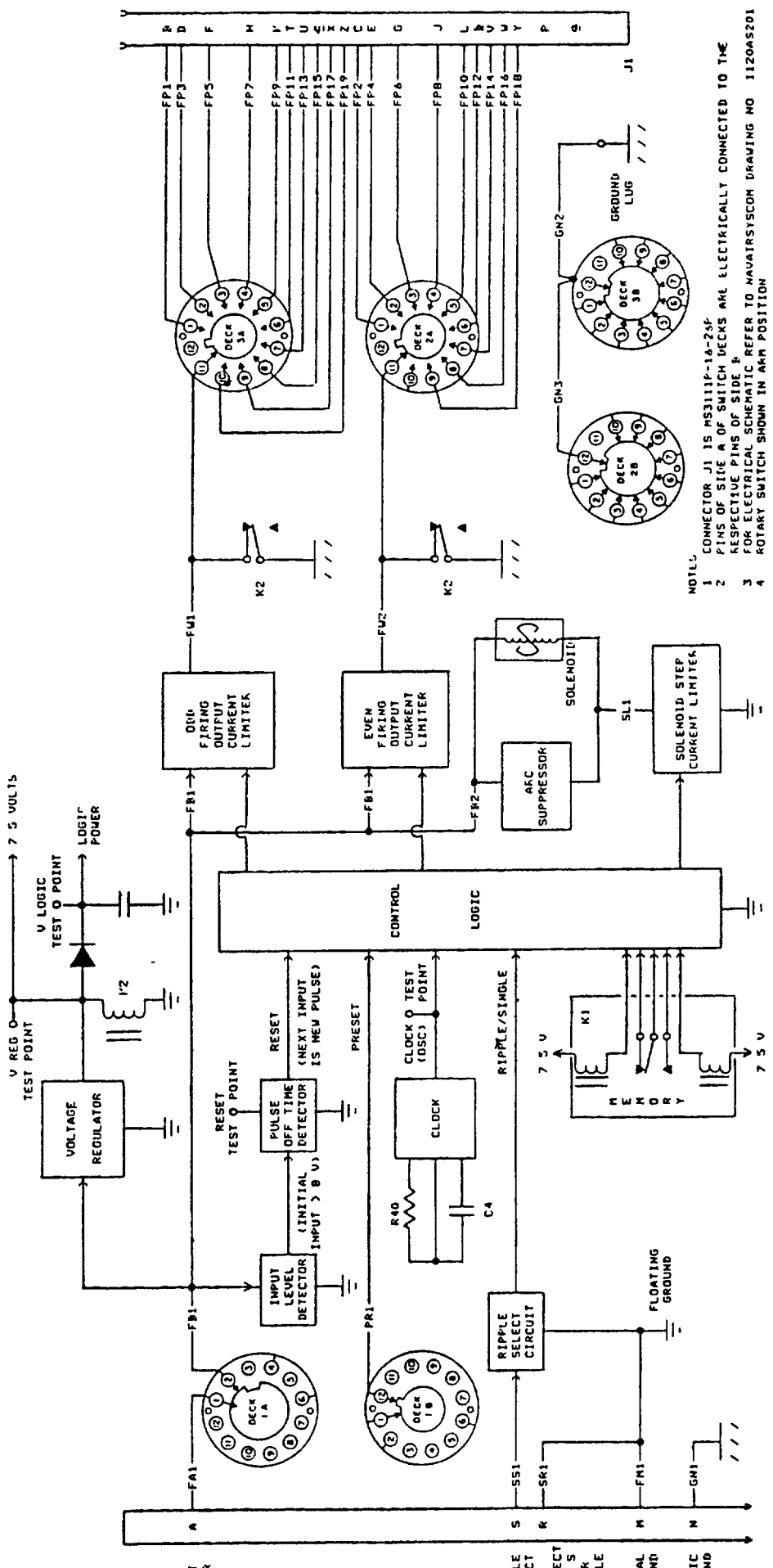


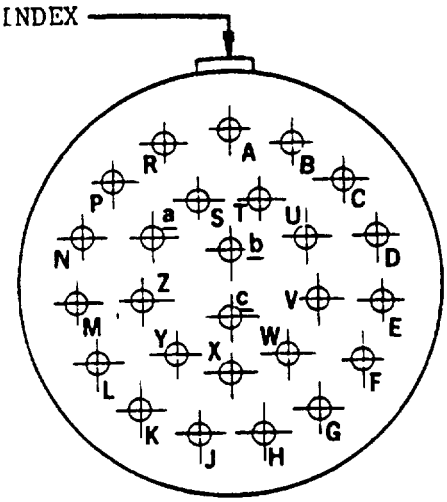

FIGURE 2. Block diagram.



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| | |
|---|---|
| INSERT ARRANGEMENT | 16-26 |
| SERVICE RATING | 1 |
| NUMBER OF CONTACTS | 26 |
| CONTACT SIZE | 20 |

CONNECTOR MATES WITH MS3116P 16 26S




PIN ARRANGEMENT

SUMMARY OF CONNECTOR INPUTS AND OUTPUTS

| FUNCTION | CONNECTOR PIN |
|---|---|
| FIRE SIGNAL OUTPUT # 1 | B |
| FIRE SIGNAL OUTPUT # 2 | C |
| FIRE SIGNAL OUTPUT # 3 | D |
| FIRE SIGNAL OUTPUT # 4 | E |
| FIRE SIGNAL OUTPUT # 5 | F |
| FIRE SIGNAL OUTPUT # 6 | G |
| FIRE SIGNAL OUTPUT # 7 | H |
| FIRE SIGNAL OUTPUT # 8 | J |
| FIRE SIGNAL OUTPUT # 9 | K |
| FIRE SIGNAL OUTPUT #10 | L |
| FIRE SIGNAL OUTPUT #11 | T |
| FIRE SIGNAL OUTPUT #12 | b |
| FIRE SIGNAL OUTPUT #13 | U |
| FIRE SIGNAL OUTPUT #14 | V |
| FIRE SIGNAL OUTPUT #15 | c |
| FIRE SIGNAL OUTPUT #16 | W |
| FIRE SIGNAL OUTPUT #17 | X |
| FIRE SIGNAL OUTPUT #18 | Y |
| FIRE SIGNAL OUTPUT #19 | Z |
| FIRE SIGNAL INPUT | A |
| RIPPLE SELECT | S |
| RIPPLE SELECT RETURN | R |
| SIGNAL GROUND (NEGATIVE RETURN) | M |
| STATIC GROUND (CASE GROUND) | N |
| NOT USED | P |
| NOT USED | a |

NOTES

1 THE UNUSED PINS SHALL HAVE A MINIMUM OF 1 MEGOHM RESISTANCE TO ANY OTHER INPUT OR OUTPUT PIN AND SHALL BE AVAILABLE FOR FUTURE USE

2 THE CONNECTOR SHALL NOT BE CONNECTED TO OR DISCONNECTED FROM THE TEST CIRCUIT WHILE POWER IS APPLIED (SWITCH S1 CLOSED)

FIGURE 3. Connector terminal identification.



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REQUIREMENTS:

Operational Data:

Function: This intervalometer, to be used in seven tube and nineteen tube 2.75 inch rocket launchers, is a reusable, nineteen (19) output, direct line operated intervalometer for which a five ohm series resistor is not required. (A detailed description of the intervalometer operation is given in 3.20 of MIL-I-85581. A block diagram is shown in Figure 2.)

Major components: The intervalometer consists of a printed wiring board assembly, an electromechanical stepping switch, and a mechanical housing as described below:

a. Printed wiring board assembly: The printed wiring board assembly consists of integrated circuits, transistors, diodes, capacitors, resistors, and relays mounted on a flexable printed wiring board. The printed wiring board assembly is defined in NAVAIRSYSCOM Drawing No. 1120AS270. Inputs and outputs to/from the printed wiring board assembly are as follows:

(1) Fire power input (wire FB1): Provides the input power to energize the logic and control functions and the current required to power the firing output switches.

(2) Firing outputs (wires FW1 and FW2): Provides a firing signal at the proper time to one of two wafers on the stepping switch. The stepping switch routes the firing signal to the proper pin on the intervalometer connector. The firing outputs are current limited and each provide an output for 15 milliseconds for alternate firing pulses.

(3) Solenoid step output (wire SL1): Provides a ground for the solenoid that steps the stepping switch to the next position. The solenoid step circuitry is current limited and provides an output (ground) for 25 milliseconds at the start of each odd firing power pulse (no output is provided during even firing power pulses).

(4) Ripple input (wire SS1): When not grounded allows the intervalometer to sequence automatically through all firing positions, at a rate of 40 milliseconds per position, until power is removed from the fire power input. Grounding the wire causes the intervalometer to operate in the single mode allowing only one output each time firing power is applied to the printed wiring board assembly.

(5) Preset circuit input (wire PR1): Receives an input which exist only when the stepping switch is in the arm position. This input causes the circuit board to be initiatized to odd pulse when the first firing power pulse is received.

(6) Power return (wire FM1): Provides a floating ground for the electronics of the printed wiring board assembly.

(7) Ripple return (wire SR1): Provides a ground for the ripple (single) select circuits.



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(8) Ground lug (wire GN4): Forms part of the static ground return circuit.

b. Stepping switch: The stepping switch is located within the intervalometer housing. This stepping switch consists of a 12 position rotary solenoid ratchet mechanism to allow any of the 12 shaft positions to be selected. The switch distributes the incoming firing signal in sequence to each of up to ten electrically actuated igniters per firing deck. The stepping switch has two (2) firing decks, and one (1) arm deck. A detailed description of the stepping switch is given in MIL-S-85242/2.

c. Mechanical housing: The mechanical housing provides a water tight enclosure for the printed wiring board assembly and the stepping switch. The housing also provides shielding from electromagnetic radiation.



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Electrical:

1. Input Voltage:

   a. Nominal: 27.5 VDC

   b. Input range: 22 to 31.5 VDC

   c. Input voltage switch contact bounce: Up to 5.0 milliseconds of contact bounce (see Figure 7).

   d. Input voltage rise time: The input voltage rise time shall be 10.0 milliseconds or less (straight line from +10 percent to +90 percent of input voltage - see Figure 7).

   e. Pickup voltage: 22 VDC maximum (5.0 ohm series resistor) over the temperature range of -55°C to +71°C.

   f. Dropout voltage: 0.5 VDC minimum (0.5 ohm series resistor) over the temperature range of -55°C to +71°C.

   g. Dropout time: Voltage dropouts less than 5.0 milliseconds shall not cause the intervalometer to sequence to the next position. Voltage dropouts longer than 15.0 milliseconds, occurring after the Fire Signal Output Delay, shall cause the intervalometer to sequence to the next position. This requirement applies over the temperature range of -55°C to +71°C. Typical dropout is illustrated in Figure 8.

2. Series input resistance: The series input resistance can range from 0.5 ohm to 7.2 ohm for input voltages from 24.5 to 31.5 volts. The resistance can vary from 0.5 ohm to 5.7 ohm for input voltages from 22.0 to 24.0 volts.

3. Series input inductance: The series input inductance can range up to 200 microhenrys.

4. Steady state input current: The steady state input current (measured after the pulse has been applied for a minimum of 50 milliseconds, single mode) shall be 0.20 ± 0.10 amperes for the full range of input voltage, line resistance, and temperature.

5. Peak input current: The peak input current shall not exceed 2.6 amperes for longer than 2.0 milliseconds.

6. Timing requirements: The timing requirements are given in Figure 1 of MIL-I-85581 and in the following paragraphs:

   a. Externally-pulsed stepping speed (single mode): The intervalometer shall be capable of following a pulse having a minimum "on" time of 40 milliseconds and a minimum "off" time of 15 milliseconds at any input voltage between 22.0 and 31.5 volts.

   b. Externally-pulsed stepping speed (ripple mode): The intervalometer shall be capable of completing its self generated firing sequence and



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return to the "safe" position in a maximum of 0.930 seconds at any input voltage between 22.0 and 31.5 volts. The time between firing pulses shall range from 37.5 milliseconds to 44.0 milliseconds.

c. Time to fire: The time to fire (Fire Signal Output Delay) shall be 25.5 ± 2.5 milliseconds (measured from application of the input pulse until the firing output occurs).

d. Internally generated contact bounce: Steady output shall occur within the interval specified by the initial delay time. No internally generated contact bounce is permitted on the firing outputs.

e. Maximum continuous "on" time: The maximum continuous "on" time shall not exceed 2 minutes in any 5 minute period.

f. Output pulse width (dwell time): The output current pulse width shall be 15 ± 2 milliseconds, assuming continuous application of input power and load.

7. Output sequence: The output sequence "firing order" is listed in Figure 3.

8. Output voltage: For testing purposes, the output voltage shall be present (after the initial delay time) for the duration of the input pulse (single mode).

9. Output (load) resistance: The output (load) resistance can range from 0.1 ohm for a shorted load to infinity for an unloaded output. The typical output load ranges from 0.7 ohm to 3.0 ohm.

10. Static ground: The static ground is separate from the negative return for the equipment and shall be capable of carrying 5 amperes current for one minute duration. The enclosure is connected to static ground.

11. Ground circuit resistance (intervalometer de-energized), any shaft position: The resistance measured between any firing pin and the static ground pin (pin N) shall not exceed 0.2 ohm.

12. Signal ground (negative return): The signal ground is the negative return for the equipment and shall be capable of carrying 5 amperes current for one minute duration. The enclosure is insulated from the signal ground.



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Physical:

1. Enclosure design: The intervalometer shall have a water tight metal case.

2. Dimensions and configuration: The intervalometer dimensions and configuration shall be as specified in Figure 1.

3. Construction: The construction of the intervalometer is specified in NAVAIRSYSCOM Drawing No. 1120AS200.

4. Finish: The intervalometer housing components shall be finished as specified in NAVAIRSYSCOM Drawing No. 1120AS200.

5. Weight: The intervalometer weight, excluding cables, shall be 0.5 ± 0.1 pound.

6. Connector: The intervalometer shall mate with the connector specified in Figure 3.

7. Identification plate: The identification plate is specified in NAVAIRSYSCOM Drawing No. 1120AS263.

8. Load/arm position indicator: The load/arm position indicator is specified in NAVAIRSYSCOM Drawing No. 1120AS246.

9. Intervalometer switch knob: The switch knob is specified in NAVAIRSYSCOM Drawing No. 1120AS248.

10. Switch knob alignment: The knob shall be installed on the switch shaft such that when the intervalometer is in the safe (load) position the switch knob index is aligned with the "L" on the load/arm plate and both set screws of the switch knob are perpendicular to the flats of the switch shaft.

Environmental:

1. Environmental conditions: The environmental conditions are specified in MIL-I-85581.

Life test requirements:

1. Operating life: The operating life shall be tested by performing 100 cycles of each of eight (8) life tests listed in Table I.



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Quality assurance inspection:

Inspection consists of: (a) first article, individual, and lot inspection on completed intervalometer, and (b) manufacturing tests on the circuit board subassembly of the intervalometer.

a. First article, individual, and lot inspection tests are conducted as specified in MIL-I-85581 by the manufacturer and such testing facilities as specified in the contract.

b. Manufacturing (circuit board) tests are required to ensure that the circuit board (NAVAIRSYSCOM Drawing No. 1120AS270) has been constructed and assembled correctly. The circuit board, tested in the flat, has several test points not available once folded. The circuit board, once folded, must remain folded. Unfolding and refolding the circuit board can damage the internal copper circuit.



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First article, individual, and lot inspection testing:

1. Electrical tests: The circuit of Figure 4 shall be used in testing the intervalometer. Table I lists the tests to be performed with required input and load parameters. Additionally, Table I lists minimum and maximum acceptable output parameters for the intervalometer under test. Figure 6 illustrates the timing relationships specified in Table I. Typical input voltage characteristics are shown in Figure 7. A typical input voltage dropout is illustrated in Figure 8. All tests in Table I shall be performed at each of the following temperatures for first article and lot inspection tests:

   a. $+71^{\circ} \pm 2^{\circ}C$

   b. $+23^{\circ} \pm 5^{\circ}C$

   c. $-55^{\circ} \pm 2^{\circ}C$

   All tests in Table I shall be performed at $23^{\circ} \pm 5^{\circ}C$ for individual inspections.

2. Environmental requirements: The intervalometer shall meet all electrical test parameters during and after being subjected to environmental testing. During the environmental test, the intervalometer shall be supported as shown in Figure 5.

3. Test sample size, for first article and lot inspection tests, shall be determined at the time acquisition, but shall be not less than nine (9) intervalometers per lot. Intervalometers used as environmental or life test samples shall not be installed in launchers. Test sample size for individual inspections shall be 100 percent.

4. Test mounting fixture: The intervalometer shall be mounted as shown in Figure 5.

5. Sequence test: The intervalometer firing output shall be checked to ensure that the output sequence is as specified in Figure 3.

6. Test voltage: The test voltage shall be as specified in Table I.

7. Safe and arm test procedures: The intervelometer shall be tested for 100 percent compliance with safe and arm circuit test requirements.

   a. Safe (load) test.

      (1) The intervalometer shall be connected to the test circuit of Figure 4 and set to the safe (load) position. The firing outputs shall not be loaded (loads (R FP) removed). Prior tc application of power, the resistance shall be measured between each of the firing output pins and ground using an ohmmeter with low ohm capability. The resistance (excluding test equipment resistance) shall not exceed 0.19 ohm.



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(2) The intervalometer shall be connected to the test circuit of Figure 4 and set to the safe (load) position. The firing outputs shall be loaded with 1.0 ± 0.1 ohm loads. Each of the firing output pins shall be monitored during the test using an oscilloscope or other suitable test instrument. When switch S1 is closed, the intervalometer shall not sequence out of the safe (load) position and the voltage monitored across each 1.0 ± 0.1 ohm load shall not exceed 50 millivolts. The two test conditions are as follows:

(a) Switch S2 set to single, V S set to 22.0 VDC and R IN set to 5.0 ± 0.1 ohm.

(b) Switch S2 set to single, V S set to 31.5 VDC and R IN set 0.5 ohm or less.

b. Arm test.

(1) The intervalometer shall be connected to the test circuit of Figure 4 and set to the arm position. The firing outputs shall not be loaded (loads (R FP) removed). Prior to application of power, the resistance shall be measured between each of the firing output pins and ground using an ohmmeter with low ohm capability. The resistance (excluding test equipment resistance) shall not exceed 0.19 ohm.

(2) The intervalometer shall be connected to the test circuit of Figure 4 and set to the arm position. The firing outputs shall not be loaded (loads (R FP) removed). Prior to application of power, the resistance shall be measured between each of the firing output pins and ground while rotating the intervalometer selector knob through 360 degrees (back to arm). An ohmmeter with low ohm capability shall be used. The resistance (excluding test equipment resistance) shall not exceed 0.19 ohm.

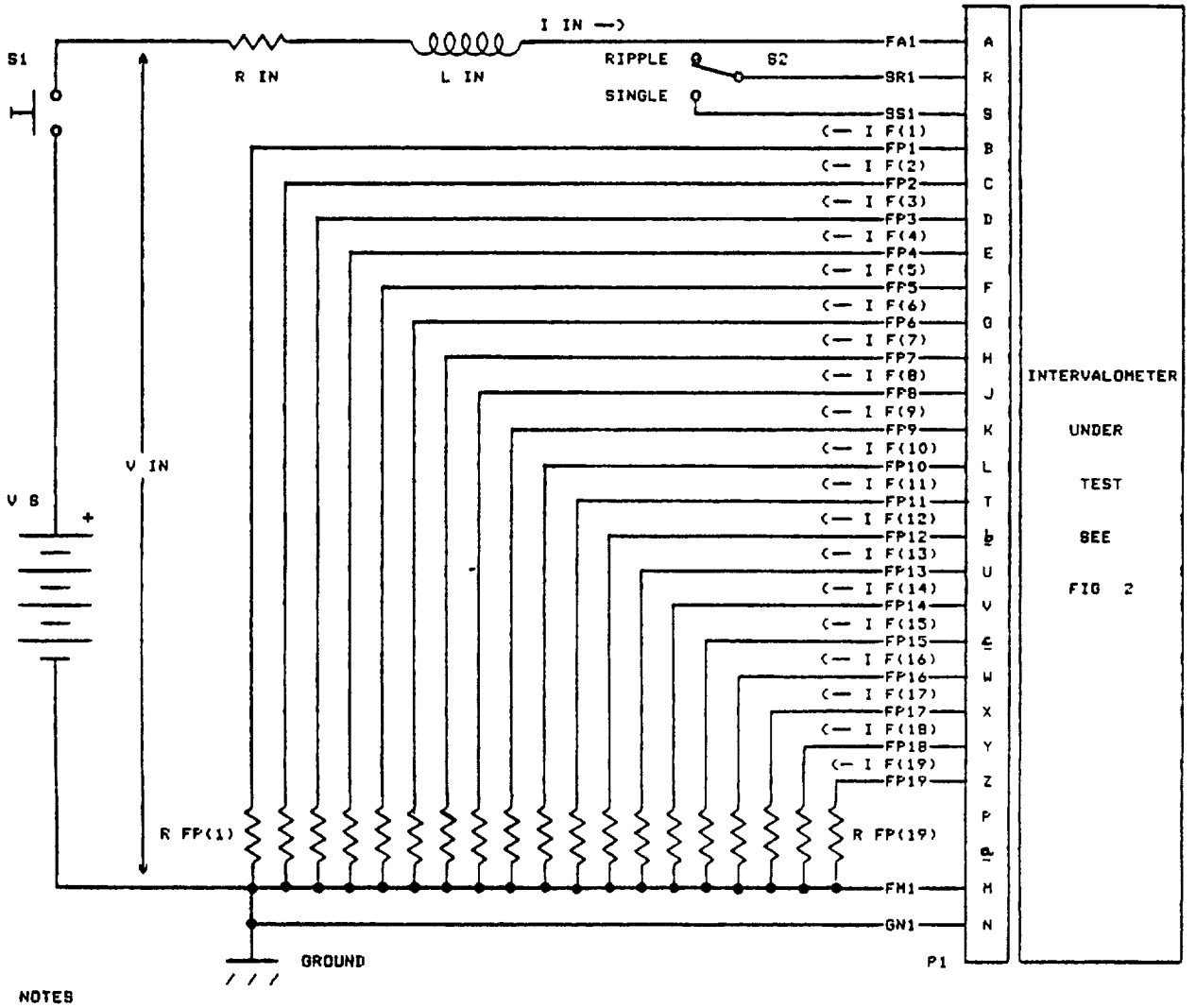
(3) The intervalometer shall be connected to the test circuit of Figure 4 and set to the arm position. The firing outputs shall be loaded with 1.0 ± 0.1 ohm loads. Each of the firing output pins shall be monitored during the test using an oscilloscope or other suitable test instrument. When switch S1 is closed, the intervalometer shall sequence out of the arm position into firing position 1. The voltage monitored across each 1.0 ± 0.1 ohm load, except firing position no. 1, shall not exceed 50 millivolts. The voltage at firing position no. 1 shall not exceed 50 millivolts for a minimum of 23 milliseconds after S1 is closed. Test conditions are as follows:

(a) Switch S2 set to single, V S set to 31.5 VDC and R IN set to 0.5 ohm or less.



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| PARTS LIST | | |
|---|---|---|
| PART NUMBER | PART NAME | REF DES |
| AS REQUIRED | POWER SOURCE (22 0 TO 31 5 VOLTS D C , 5 AMP ) | V S |
| AS REQUIRED | SWITCH (SPST, NORMALLY OPEN, 5 AMP CONTACTS) | S1 |
| AS REQUIRED | SWITCH (SPST, 0 5 AMP CONTACTS) | S2 |
| MS3116P-16-26S | CONNECTOR (26 PINS) | P1 |
| INPUT RESISTANCE - SPECIFIED IN TABLE I | | R IN |
| LOAD RESISTANCE - SPECIFIED IN TABLE I | | R FP(1 - 19) |
| INPUT INDUCTANCE - SPECIFIED IN TABLE I | | L IN |

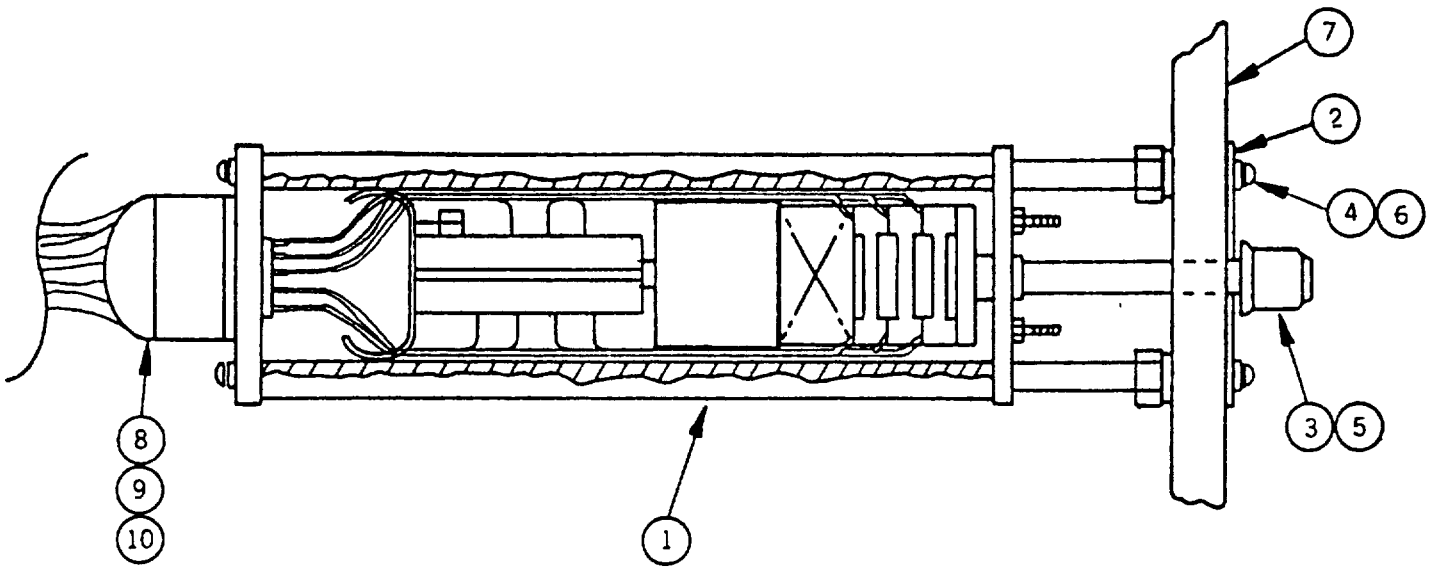



NOTES
1 R IN INCLUDES LINE RESISTOR, WIRE RESISTANCE, SWITCH CONTACT RESISTANCE, POWER SOURCE RESISTANCE, AND TEST EQUIPMENT RESISTANCE
2 R FP (1 THRU 19) INCLUDES LOAD RESISTOR, WIRE RESISTANCE, SWITCH CONTACT RESISTANCE, AND TEST EQUIPMENT RESISTANCE
3 L IN INCLUDES LINE INDUCTOR, WIRE INDUCTANCE, POWER SOURCE INDUCTANCE, AND TEST EQUIPMENT INDUCTANCE
4 SWITCHES S1 AND S2 MAY BE REPLACED BY RELAYS AND/OR SOLID STATE DEVICES

FIGURE 4. Electrical test circuit for first article, individual, and lot inspection testing.



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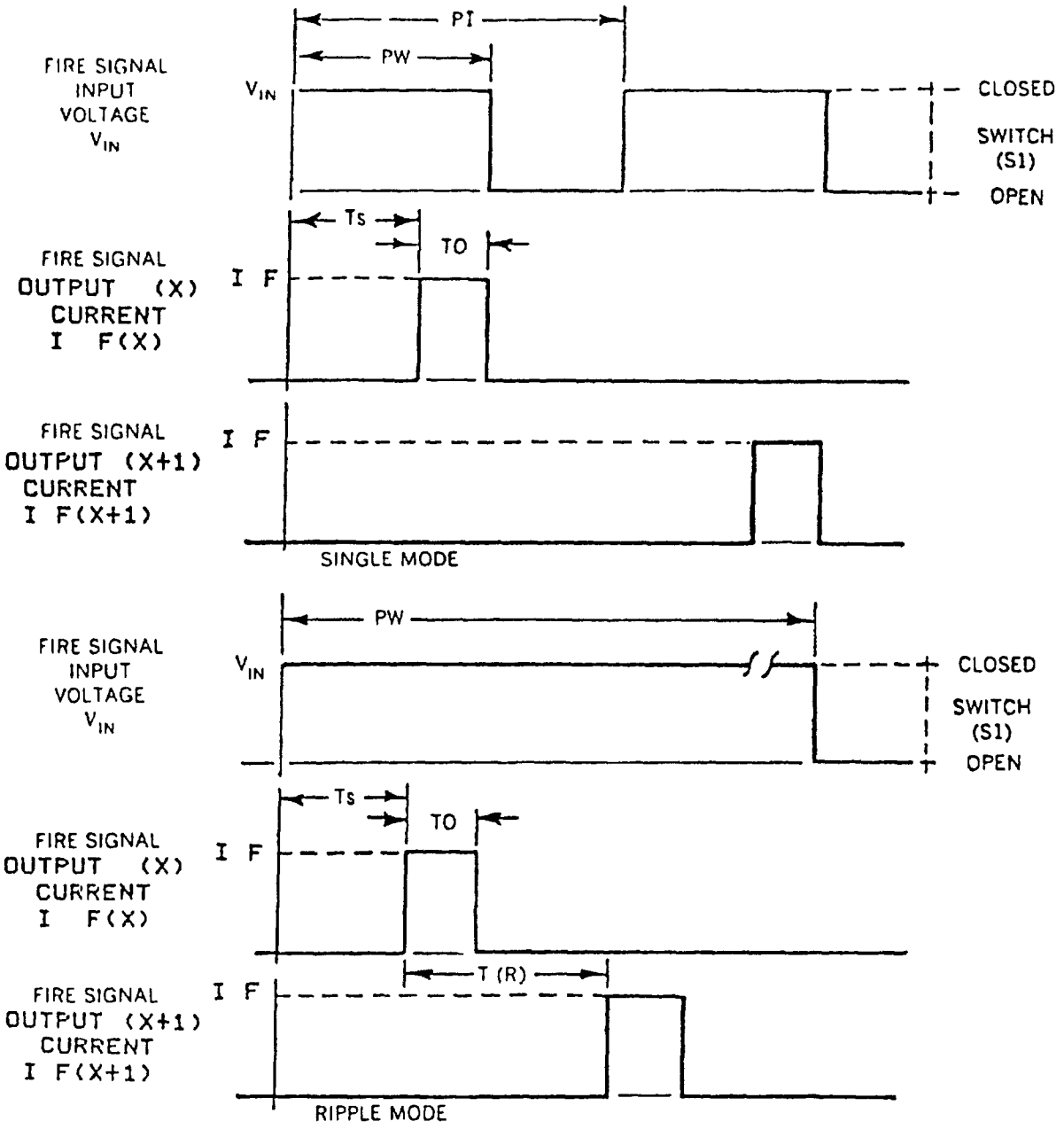

PARTS LIST

| REQD | PART NUMBER | PART NAME | ITEM |
|---|---|---|---|
| 1 | M85581/1-1 | INTERVALOMETER | 1 |
| 1 | 1120AS246 | PLATE, LOAD/ARM | 2 |
| 1 | 1120AS248 | KNOB, SWITCH | 3 |
| 3 | MS35207-231 | SCREW | 4 |
| 2 | MS51021-101 | SCREW, SET | 5 |
| 3 | MS35335-30 | WASHER | 6 |
| 1 | FIGURE 9 | MOUNTING BULKHEAD | 7 |
| 1 | MS3116F-16-26S | MATING CONNECTOR | 8 |
| AR | QQ-S-571, SN60 OR SN63 | SOLDER | 9 |
| AR | 200AS108-22C10 | WIRE | 10 |

FIGURE 5. Intervalometer mounting for test.



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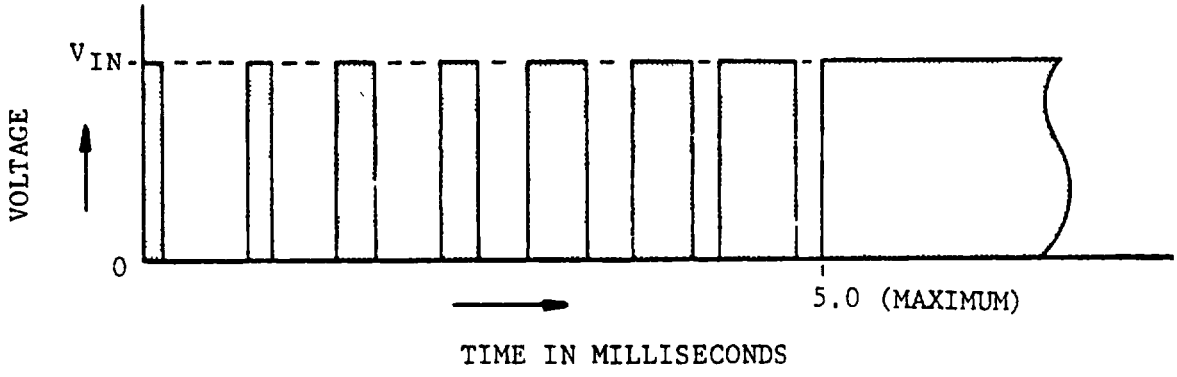



NOTES
1. OUTPUT (X) IS ANY OUTPUT 1 THRU 19
2. FIRE SIGNAL OUTPUT DELAY (TIME TO FIRE), T(S), IS 25 MILLISECONDS
3 OUTPUT FIRE CURRENT PULSE WIDTH (DWELL TIME), T(O), IS 15 MILLISECONDS (CONSTANT LOAD)
4 RIPPLE INTERVAL (T (R)), MEASURED FROM THE LEADING EDGE OF ONE FIRING PULSE TO THE LEADING EDGE OF THE NEXT FIRING PULSE, IS 41 MILLISECONDS.

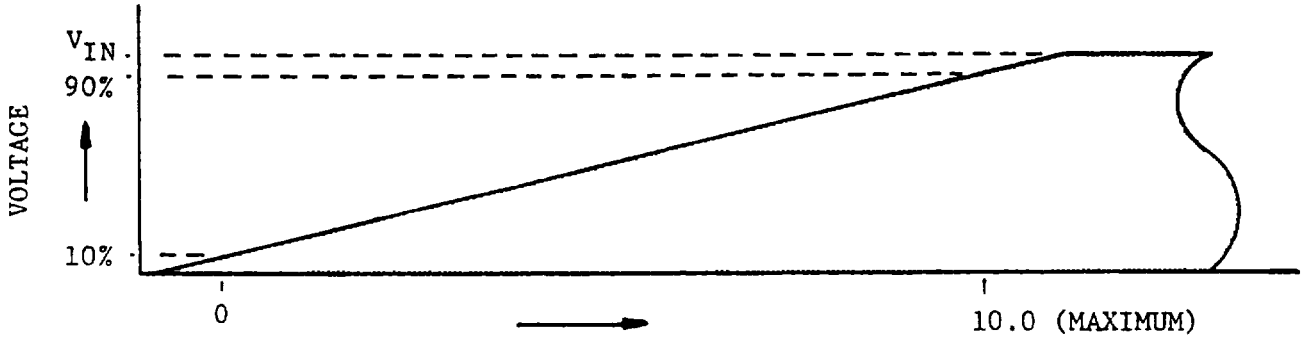
FIGURE 6. Intervalometer timing relationship.



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TYPICAL INPUT SWITCH CONTACT BOUNCE

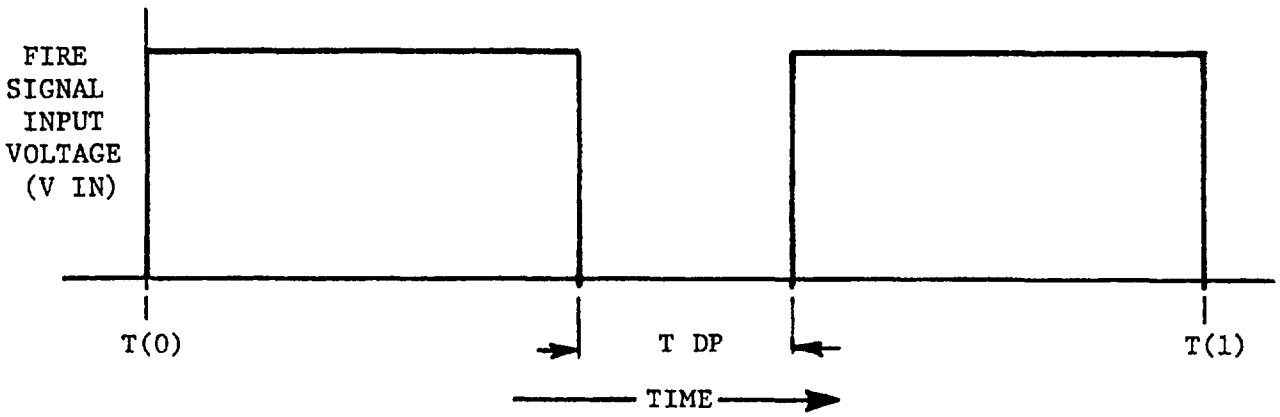



TYPICAL INPUT SLOW RISE

FIGURE 7. Input voltage characteristics.



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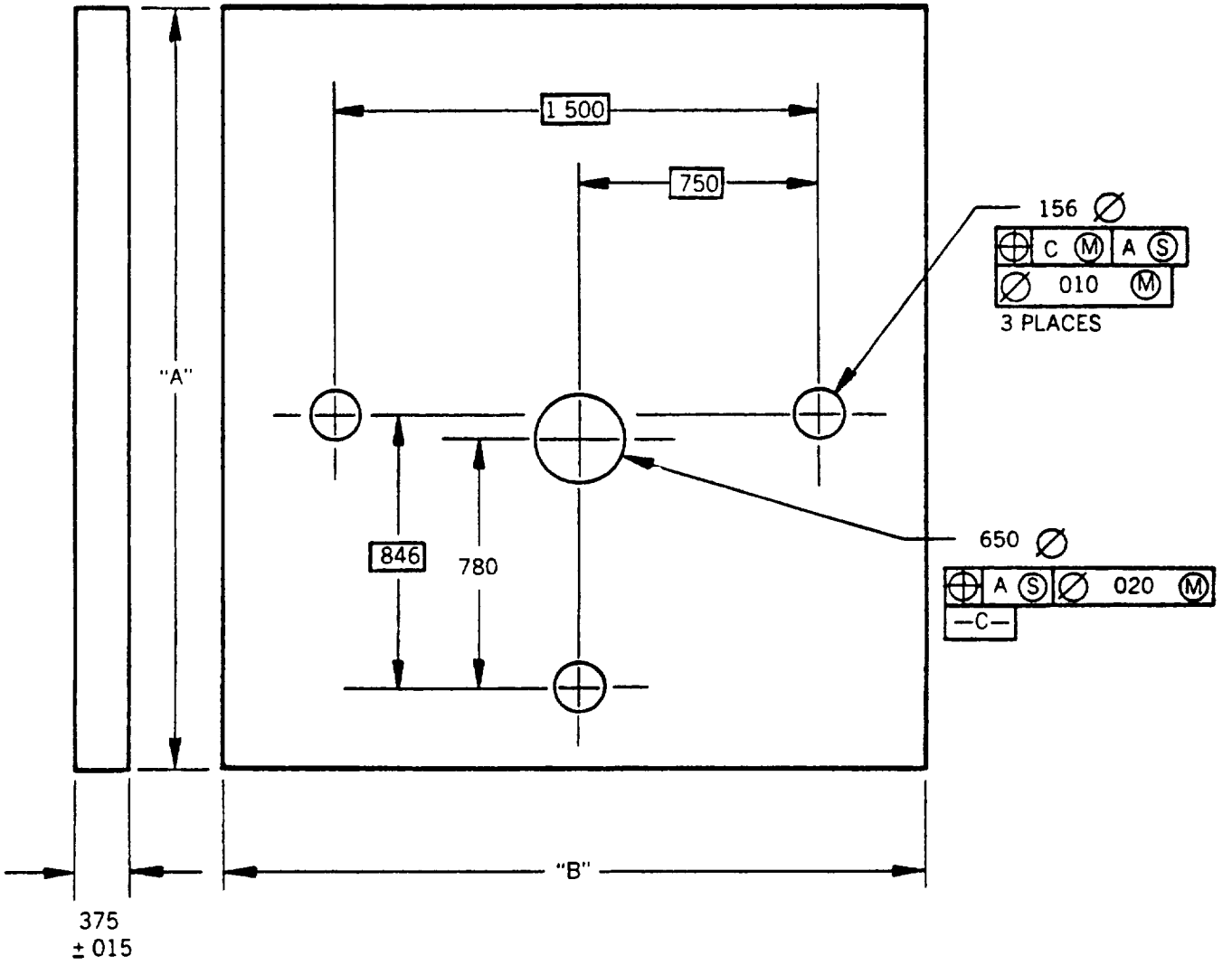

NOTES:

1. FIRE SIGNAL STARTS AT T(0).
2. FIRE SIGNAL ENDS AT T(1).
3. DROPOUT (T DP) LESS THAN 5.0 MILLISECONDS SHALL NOT CAUSE THE INTERVALOMETER TO ADVANCE TO NEXT POSITION.
4. DROPOUT (T DP) GREATER THAN 15.0 MILLISECONDS OCCURRING AFTER THE FIRE SIGNAL OUTPUT DELAY SHALL CAUSE THE INTERVALOMETER TO ADVANCE TO NEXT POSITION.
5. DROPOUT CAN OCCUR AT ANY TIME DURING DURATION OF FIRE SIGNAL.

FIGURE 8. Typical dropout characteristics.



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NOTES

1 MATERIAL ALUMINUM
2 DIMENSIONS "A" AND "B" AS REQUIRED.

FIGURE 9. Mounting bulkhead.



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# TABLE I. Intervalometer electrical test parameters.

| TEST NO | TEST TYPE | INPUT: INPUT VOLTAGE V IN VOLTS | INPUT: LINE RESISTANCE R IN OHMS | INPUT: LINE INDUCTANCE L IN MICROH | INPUT: PULSE WIDTH PW MS | INPUT: PULSE INTERVAL PI MS | INPUT: PULSE QTY | LOAD: FIRE OUTPUT RESISTANCE R FP(1 - 19) OHMS | RESPONSE PARAMETERS: FIRE OUTPUT (OUTPUTS 1 THRU 19) | | | | | MILSPEC TEST REFERENCE | | | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | FIRE CURRENT I F | | TIME TO FIRE T S | | DWELL T O | LIFE TEST 4 8 | SIMULATED FIRING TEST 4 9 1 | SEQUENCE TEST 4 9.5 | BYPASS TEST 4 9 6 | OPERATIONAL TEST 4 9 7 | |
| | | | | | | | | | MINIMUM AMP | MAXIMUM AMP | MINIMUM MS | MAXIMUM MS | MINIMUM MS | | | | | | |
| | | NOTE 4 | NOTE 6 | NOTE 8 | | | | NOTES 5 & 7 | NOTE 14 | | | | NOTE 11 | NOTE 15 | | | | | NOTES 1 THRU 15 APPLY ALL TESTS |
| I-1 | SEQUENCE | 22.0 | 5 7 | 40 MAXIMUM | 400 MINIMUM | 550 MINIMUM | 21 | 1 1 | 1 25 | 1 95 | 23 0 | 28 0 | 10 5 | X | | X | | X | |
| I-2 | PULSE (MINIMUM) | 31 5 | 0 5 | 150 | 40 | 56 | 21 | 1 1 | 1 65 | 2 45 | 23 0 | 28 0 | 10 5 | X | | | | X | |
| I-3 | PULSE (MINIMUM) | 31 5 | 0 5 | 150 | 40 | 56 | 21 | 2 5 | 1 65 | 2 45 | 23 0 | 28 0 | 10 5 | | | | | X | |
| I-4 | PULSE (MINIMUM) | 22 0 | 5 7 | 150 | 40 | 56 | 21 | 1 1 | 1 25 | 1 95 | 23 0 | 28 0 | 10 5 | | | | | X | |
| I-5 | PULSE (MINIMUM) | 22 0 | 5 7 | 150 | 40 | 56 | 21 | 2 5 | 1 20 | 1 95 | 23 0 | 28 0 | 10 5 | | | | | X | |
| I-6 | PULSE NOTE 16 (SWITCH NOISE) | 31 5 | 0 5 | 150 | 45 +5 NOISE | 68 | 21 | 1 1 | 1 65 | 2 45 | 23 0 | 33 0 | 10 5 | X | X | | | X | |
| I-7 | PULSE NOTE 16 (SWITCH NOISE) | 22 0 | 5 7 | 150 | 45 +5 NOISE | 68 | 21 | 1 1 | 1 25 | 1 95 | 23 0 | 33 0 | 10 5 | | X | | | X | |
| I-8 | PULSE NOTE 17 (SLOW RISE) | 31 5 | 0 5 | 150 | 45 +5 RAMP | 68 | 21 | 1 1 | 1 65 | 2 45 | 24 0 | 30 0 | 10 5 | | | | | X | |
| I-9 | PULSE NOTE 17 (SLOW RISE) | 22 0 | 5 7 | 150 | 45 +5 RAMP | 68 | 21 | 1 1 | 1 25 | 1 95 | 24 5 | 32 0 | 10 5 | X | | | | X | |
| I-10 | PULSE NOTE 20 (DROPOUT) | 22 0 | 5 7 | 150 | 48 -4 DROPOUT | 68 | 21 | 1 1 | 1 25 | 1 95 | 23 0 | 32 0 | 10 5 | | | | | X | NOTES 18, 19, & 21 |
| I-11 | PULSE (BYPASS) | 31 5 | 0 5 | 150 | 40 | 56 | 21 | SHORT, OPEN, OR 1 1 OHM | 1 65 | 2 45 | 23 0 | 28 0 | 10 5 NOTE 12 | X | | | X | X | NOTES 22 & 23 |
| I-12 | PULSE (BYPASS) | 22 0 | 5 7 | 150 | 40 | 56 | 21 | SHORT, OPEN, OR 1 1 OHM | 1 25 | 1 95 | 23 0 | 28 0 | 10 5 NOTE 12 | | | | X | X | NOTES 22 & 23 |
| I-13 | RIPPLE | 27 5 | 5 7 | 150 | 930 | 10K MINIMUM | 1 | 1 1 | 1 35 | 2 20 | 23 0 | 28 0 | 10 5 | X | | | | X | NOTE 24 |
| I-14 | RIPPLE (BYPASS) | 31 5 | 0 5 | 150 | 930 | 10K MINIMUM | 1 | SHORT, OPEN, OR 1 1 OHM | 1 65 | 2 45 | 23 0 | 28 0 | 10 5 NOTE 12 | X | X | | X | X | NOTES 22, 23, & 24 |
| I-15 | RIPPLE (BYPASS) | 22 0 | 5 7 | 150 | 930 | 10K MINIMUM | 1 | SHORT, OPEN, OR 1 1 OHM | 1 25 | 1 95 | 23 0 | 28 0 | 10 5 NOTE 12 | X | X | | X | X | NOTES 22, 23, & 24 |





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TABLE I. Intervalometer electrical test parameters. - Continued

Notes:

1. Abbreviations: MS - milliseconds; AMP - amperes; MA - milliamperes; QTY - quantity; MICROH - microhenry.
2. Tolerance: Resistance ± 0.1 ohm; voltage ± 1 percent; time ± 2 percent; current ± 2 percent; inductance ± 20 percent.
3. Temperature range: -55° to +71°C, unless otherwise specified.
4. Input voltage rise time: 0.1 milliseconds or less (except tests no. I-8 and I-9).
5. Remove 1.1 or 2.5 ohm load after 12 ± 1.5 milliseconds, final load resistance shall be greater than 100K ohms.
6. Input line resistance includes line resistor, wire resistance, switch contact resistance, power source resistance, and test equipment resistance.
7. Fire load resistance includes load resistor, wire resistance, switch contact resistance, and test equipment resistance.
8. Input line inductance includes line inductor, wire inductance, power source inductance, and test equipment inductance.
9. Set intervalometer switch knob to arm "A" prior to the start of each test.
10. The intervalometer switch knob shall stop at the safe/load "L" position at the end of each test.
11. Maximum fire output dwell is measured only when output load is not removed and shall not exceed 17.0 milliseconds. Minimum dwell time for a continuous load is 13.5 milliseconds.
12. Fire current "dwell time" (T O) does not apply to unloaded stations.
13. The intervalometer shall not auto-step and shall not skip any positions. Auto-step is defined as intervalometer advancing to the next position even though fire signal input voltage (V IN) remains valid (Tests I-1 through I-12 only).
14. Firing current (I L) leakage on the fire output lines shall not exceed 50 milliamperes.
15. The tests from MIL-I-85581 to be performed are indicated by an "X".
16. Typical switch noise (contact bounce) is shown in Figure 7.
17. Typical slow rise input is shown in Figure 7. For this test, rise time is 5.0 ± 0.2 MS.
18. A 4 millisecond dropout can effect fire output "time to fire" and "dwell time".
19. A 4 millisecond dropout shall not cause the intervalometer to reset itself (advance to the next position) as would a new pulse.
20. Typical dropout is shown in Figure 8. For this test drop time is 4.0 ± 0.1 MS.
21. Dropouts shall occur at 7 different times (minimum), with a maximum of one dropout per pulse.
22. A shorted fire output shall be 0.1 ohms and an unloaded output shall be greater than 100K ohms.
23. Short two odd and two even outputs. Unload two odd and two even outputs. Load the remaining outputs with 1.1 ohm.
24. Ripple interval (Tests I-13 through I-15) shall range between 37.5 milliseconds (minimum) and 44.0 milliseconds (maximum).



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Manufacturing (circuit board) test:

1. Manufacturing (circuit board) tests: Required to ensure that the circuit board (NAVAIRSYSCOM Drawing No. 1120AS270) has been constructed and assembled correctly. The circuit board is tested in the flat.

2. Handling: Once folded, the circuit board must remain folded. Unfolding and refolding the circuit board can damage the internal copper circuit.

3. Electrical tests: The electrical test circuit of Figure 10 shall be used in testing the circuit board. The circuit board shall be mounted as required. Table II lists the tests to be performed with required input and load parameters and the acceptable response parameters for the circuit board under test. Figure 11 illustrates the timing relationships specified in Table II. All tests in Table II shall be performed at $23^{\circ} \pm 5^{\circ}C$.

4. Test sample size shall be 100 percent.

5. Sequence test: The circuit board step and firing outputs shall be checked to ensure that the output sequence is as specified in Figure 11.

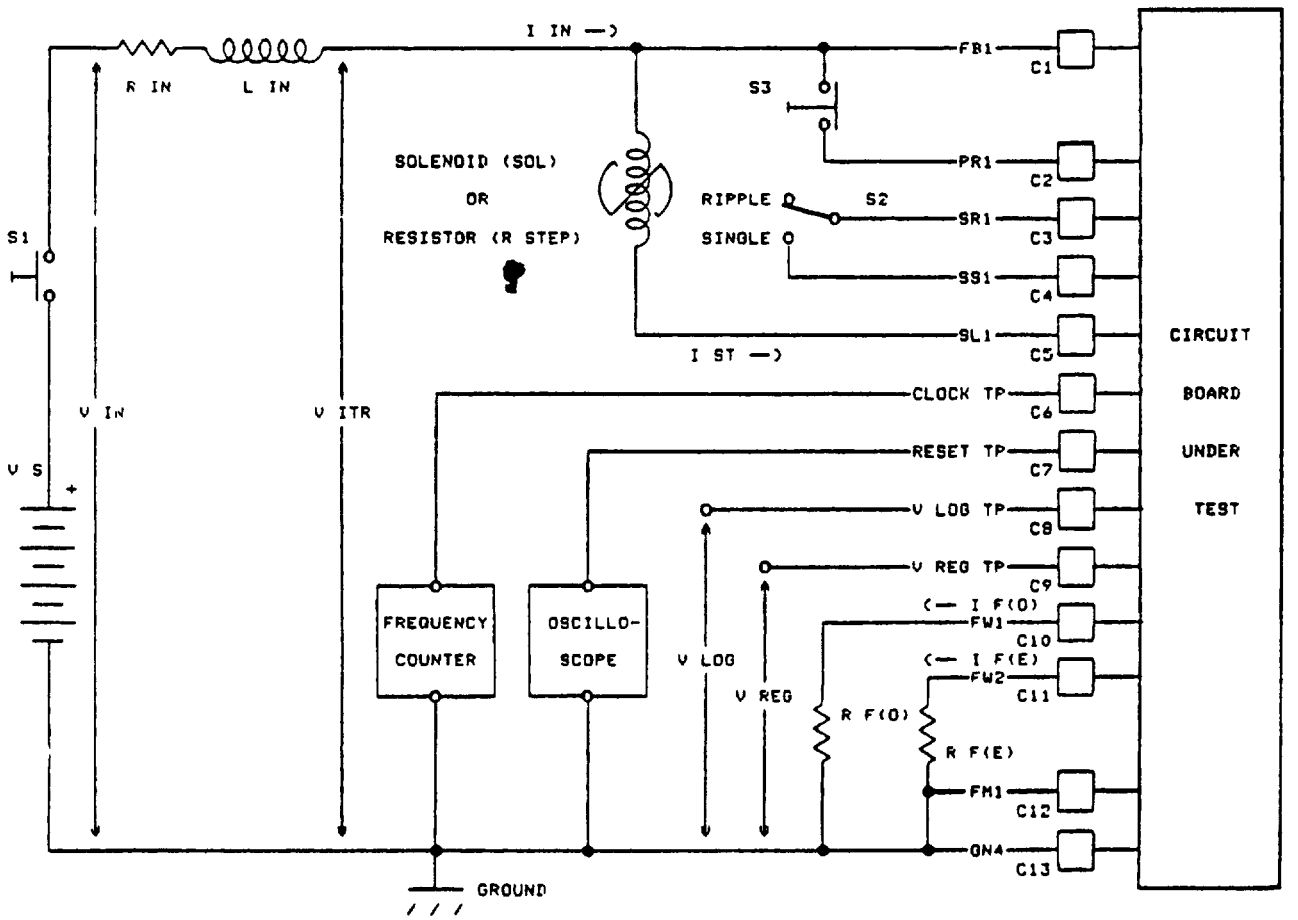
6. Test voltage: The test voltage shall be as specified in Table II.

7. Ground circuit test: With the circuit board disconnected from the test circuit, the odd and even firing output circuit resistance shall be measured. The resistance (excluding test equipment resistance), measured between wires "FW1" and "GN4" for the odd circuit and between wires "FW2" and "GN4" for the even circuit, shall not exceed 0.18 ohm.



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| PARTS LIST | | |
|---|---|---|
| PART NUMBER | PART NAME | REF DES |
| AS REQUIRED | POWER SOURCE (22.0 TO 31.5 VOLTS D.C., 5 AMP.) | V S |
| AS REQUIRED | SWITCH (SPST, NORMALLY OPEN, 5 AMP CONTACTS) | S1 |
| AS REQUIRED | SWITCH (SPST, 0.5 AMP CONTACTS) | S2 |
| AS REQUIRED | SWITCH (SPST, NORMALLY CLOSED, 0.5 AMP CONTACTS) | S3 |
| AS REQUIRED | CIRCUIT BOARD CONNECTORS | C1 THRU C13 |
| INPUT RESISTANCE - SPECIFIED IN TABLE II | | R IN |
| ODD LOAD RESISTANCE - SPECIFIED IN TABLE II | | R F(O) |
| EVEN LOAD RESISTANCE - SPECIFIED IN TABLE II | | R F(E) |
| STEP LOAD RESISTANCE AND TYPE - SPECIFIED IN TABLE II | | SOL/R STEP |
| INPUT INDUCTANCE - SPECIFIED IN TABLE II | | L IN |

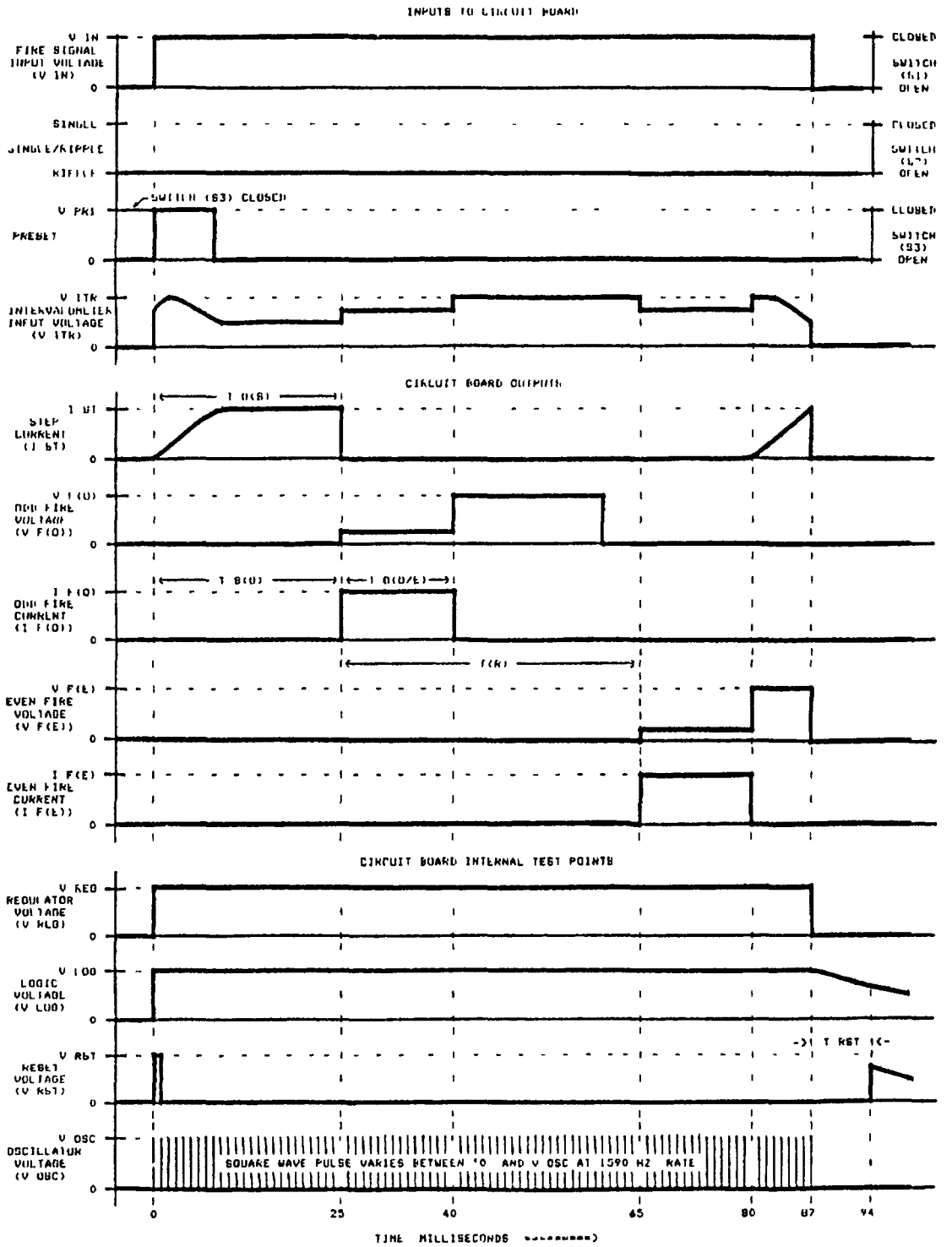



NOTES
1. R IN INCLUDES LINE RESISTOR, WIRE RESISTANCE, SWITCH CONTACT RESISTANCE, POWER SOURCE RESISTANCE, AND TEST EQUIPMENT RESISTANCE
2. R F(O) AND R F(E) INCLUDES LOAD RESISTOR, WIRE RESISTANCE, SWITCH CONTACT RESISTANCE, AND TEST EQUIPMENT RESISTANCE
3. L IN INCLUDES LINE INDUCTOR, WIRE INDUCTANCE, POWER SOURCE INDUCTANCE, AND TEST EQUIPMENT INDUCTANCE
4. SWITCHES S1, S2, AND S3 MAY BE REPLACED BY RELAYS AND/OR SOLID STATE DEVICES
5. TEST CIRCUIT CONNECTORS C6, C7, C8, AND C9 CONNECT DIRECTLY TO COMPONENTS ON THE CIRCUIT BOARD

FIGURE 10. Electrical test circuit for circuit board tests.



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NOTES
1. FIRE SIGNAL OUTPUT DELAY (ODD FIRING PULSE - TIME TO FIRE) T B(O) IS 25 MILLISECONDS
2. FIRING CURRENT PULSE WIDTH (DWELL TIME) T D(O/E) IS 15 MILLISECONDS (CONSTANT LOAD)
3. STEP CURRENT PULSE WIDTH (DWELL TIME) T B(S) IS 25 MILLISECONDS
4. RIPPLE INTERVAL T(R), MEASURED FROM THE LEADING EDGE OF ONE FIRING PULSE TO THE LEADING EDGE OF THE NEXT FIRING PULSE, IS 40 MILLISECONDS
5. SEQUENCE
   (A) RIPPLE, FIRST PULSE - STEP OCCURS FOR 25 MS , THEN FIRE ODD FOR 15 MS , THAN NOTHING FOR 25 MS THEN FIRE EVEN FOR 15 MS THEN BACK TO STEP SEQUENCE CONTINUES UNTIL POWER IS REMOVED
   (B) SINGLE, FIRST PULSE - STEP OCCURS FOR 25 MS THEN FIRE ODD FOR 15 MS THEN NOTHING AS LONG AS POWER IS APPLIED
   SECOND PULSE - NOTHING FOR 25 MS , THEN FIRE EVEN FOR 15 MS THEN NOTHING AS LONG AS POWER IS APPLIED
   THIRD PULSE - SAME AS FOR FIRST PULSE

FIGURE 11. Typical timing relationships for circuit board tests (ripple mode, constant load, shown).



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# TABLE II. Circuit board electrical test parameters.

| TEST NO | TEST TYPE | INPUT VOLTAGE V IN VOLTS (NOTE 4) | INPUT RESISTANCE R IN OHMS (NOTE 6) | INPUT INDUCTANCE L IN MICRON (NOTE 9) | PULSE WIDTH PW MS | PULSE INTERVAL PI MS | QTY | LOAD FIRE OUTPUT RESISTANCE R F(O/E) OHMS (NOTES 5 & 7) | LOAD STEP SOLENOID SOL (NOTE 8) | LOAD RESISTOR R STEP OHMS (NOTE 7) | RESPONSE PARAMETERS: FIRE OUTPUT CURRENT I F(O/E) MINIMUM AMP (NOTE 14) | FIRE OUTPUT CURRENT I F(O/E) MAXIMUM AMP (NOTE 14) | FIRE OUTPUT TIME TO FIRE T S(O/E) MINIMUM MS (NOTES 12 & 13) | FIRE OUTPUT TIME TO FIRE T S(O/E) MAXIMUM MS (NOTES 12 & 13) | FIRE OUTPUT DWELL TIME T D(O/E) MINIMUM MS (NOTE 10) | STEP OUTPUT CURRENT I ST MINIMUM AMP (NOTE 15) | STEP OUTPUT CURRENT I ST MAXIMUM AMP (NOTE 15) | STEP OUTPUT TIME TO STEP T S(ST) MAXIMUM MS | STEP OUTPUT DWELL TIME T D(S) MINIMUM MS | STEP OUTPUT DWELL TIME T D(S) MAXIMUM MS | INTERNAL VOLTAGE REGULATOR V REG MINIMUM VOLTS (NOTE 19) | LOGIC V LOG MINIMUM VOLTS (NOTE 20) | REMARKS (NOTES 1 THRU 21 APPLY ALL TESTS) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CB-1A | MANUAL STEP | 27 5 | 5 7 | 40 MAXIMUM | 200 MINIMUM | 500 MINIMUM | 1 | 1 1 | | 6 0 | 1 50 I F(O) | 2 10 I F(O) | 24 0 | 27 0 | 10 5 | 1 65 | 2 30 | 1 0 | 24 0 | 26 0 | 6 5 | 6 0 | SEE NOTES 22 & 34 |
| CB-1B | MANUAL STEP | 22 0 | 5 7 | 40 MAXIMUM | 200 MINIMUM | 500 MINIMUM | 1 | 1 1 | | 6 0 | 1 40 I F(E) | 1 85 I F(E) | 24 0 | 27 0 | 10 5 | NONE | | NA | NONE | | 6 0 | 5 5 | SEE NOTES 22 & 34 |
| CB-1C | MANUAL STEP | 31 5 | 0 5 | 150 | 200 MINIMUM | 500 MINIMUM | 1 | 1 1 | SOL | | 1 80 I F(O) | 2 35 I F(O) | 24 0 | 27 0 | 10 5 | 1 70 | 2 50 | 1 0 | 24 0 | 26 0 | 7 0 | 6 5 | SEE NOTES 22 & 34 |
| CB-1D | MANUAL STEP | 31 5 | 0 5 | 40 MAXIMUM | 200 MINIMUM | 500 MINIMUM | 1 | 1 1 | | 6 0 | 1 80 I F(E) | 2 35 I F(E) | 24 0 | 27 0 | 10 5 | NONE | | NA | NONE | | 7 0 | 6 5 | SEE NOTE 34 |
| CB-2 | PULSE (MEDIUM) | 31 5 | 0 5 | 40 MAXIMUM | 50 | 100 | 21 | 1 1 | | 6 0 | 1 80 | 2 35 | 24 0 | 27 0 | 10 5 | 1 70 | 2 50 | 1 0 | 24 0 | 26 0 | 7 0 | 6 5 | |
| CB-3 | PULSE (MINIMUM) | 22 0 | 5 7 | 40 MAXIMUM | 40 | 56 | 21 | 1 1 | | 6 0 | 1 40 | 1 85 | 24 0 | 27 0 | 10 5 | 1 40 | 1 75 | 1 0 | 24 0 | 26 0 | 6 0 | 5 5 | |
| CB-4 | PULSE NOTE 23 (SWITCH NOISE) | 31 5 | 0 5 | 150 | 40 +5 NOISE | 61 | 21 | 1 1 | SOL | | 1 80 | 2 35 | 24 0 | 32 0 | 10 5 | 1 70 | 2 50 | 5 0 | 24 0 | 31 0 | 7 0 | 6 5 | |
| CB-5 | PULSE NOTE 23 (SWITCH NOISE) | 22 0 | 5 7 | 150 | 40 +5 NOISE | 61 | 21 | 1 1 | SOL | | 1 40 | 1 85 | 24 0 | 32 0 | 10 5 | 1 40 | 1 85 | 5 0 | 24 0 | 31 0 | 6 0 | 5 5 | |
| CB-6 | PULSE NOTE 24 (SLOW RISE) | 31 5 | 0 5 | 40 MAXIMUM | 40 +5 RAMP | 61 | 21 | 1 1 | | 6 0 | 1 80 | 2 35 | 25 0 | 29 0 | 10 5 | 1 70 | 2 50 | 3 0 | 25 0 | 28 0 | 7 0 | 6 5 | |
| CB-7 | PULSE NOTE 24 (SLOW RISE) | 22 0 | 5 7 | 40 MAXIMUM | 40 +5 RAMP | 61 | 21 | 1 1 | | 6 0 | 1 40 | 1 85 | 25 5 | 31 0 | 10 5 | 1 40 | 1 75 | 3 5 | 25 5 | 30 0 | 6 0 | 5 5 | |
| CB-8 | PULSE NOTE 27 (DROPOUT) | 22 0 | 5 7 | 150 | 44 -4 DROPOUT | 60 | 21 | 1 1 | SOL | | 1 40 | 1 85 | 24 0 | 31 0 | 10 5 | 1 40 | 1 85 | 5 0 | 24 0 | 30 0 | 6 0 | 5 5 | SEE NOTES 25, 26, & 28 |
| CB-9 | PULSE (BYPASS) | 31 5 | 0 5 | 150 | 40 | 56 | 21 | SHORT, OPEN OR 1 1 OHM | SOL | | 1 80 | 2 35 | 24 0 | 27 0 | 10 5 (NOTE 11) | 1 70 | 2 50 | 1 0 | 24 0 | 26 0 | 7 0 | 6 5 | SEE NOTES 29 & 30 |
| CB-10 | PULSE (BYPASS) | 31 5 | 0 5 | 150 | 40 | 56 | 21 | SHORT OPEN OR 2 5 OHM | SOL | | 1 80 | 2 35 | 24 0 | 27 0 | 10 5 (NOTE 11) | 1 70 | 2 50 | 1 0 | 24 0 | 26 0 | 7 0 | 6 5 | SEE NOTES 29 & 31 |
| CB-11 | PULSE (BYPASS) | 22 0 | 5 7 | 150 | 40 | 56 | 21 | SHORT, OPEN OR 1 1 OHM | SOL | | 1 40 | 1 85 | 24 0 | 27 0 | 10 5 (NOTE 11) | 1 40 | 1 85 | 1 0 | 24 0 | 26 0 | 6 0 | 5 5 | SEE NOTES 29 & 30 |
| CB-12 | PULSE (BYPASS) | 22 0 | 5 7 | 150 | 40 | 56 | 21 | SHORT OPEN, OR 2 5 OHM | SOL | | 1 35 | 1 85 | 24 0 | 27 0 | 10 5 (NOTE 11) | 1 40 | 1 85 | 1 0 | 24 0 | 26 0 | 6 0 | 5 5 | SEE NOTES 29 & 31 |
| CB-13 | PULSE (PRESET) | 27 5 | 5 7 | 150 | 40 | 56 | 21 | 1 1 | SOL | | 1 50 | 2 10 | 24 0 | 27 0 | 10 5 | 1 65 | 2 30 | 1 0 | 24 0 | 26 0 | 6 5 | 6 0 | SEE NOTE 32 |
| CB-14 | RIPPLE | 27 5 | 5 7 | 150 | 930 | 10K MINIMUM | 1 | 1 1 | SOL | | 1 50 | 2 10 | 24 0 | 27 0 | 10 5 | 1 65 | 2 30 | 1 0 | 24 0 | 26 0 | 6 5 | 6 0 | SEE NOTE 33 |
| CB-15 | RIPPLE (BYPASS) | 31 5 | 0 5 | 150 | 930 | 10K MINIMUM | 1 | SHORT, OPEN, OR 1 1 OHM | SOL | | 1 80 | 2 35 | 24 0 | 27 0 | 10 5 (NOTE 11) | 1 70 | 2 50 | 1 0 | 24 0 | 26 0 | 7 0 | 6 5 | SEE NOTES 29, 30 & 33 |
| CB-16 | RIPPLE (BYPASS) | 31 5 | 0 5 | 150 | 930 | 10K MINIMUM | 1 | SHORT, OPEN, OR 2 5 OHM | SOL | | 1 80 | 2 35 | 24 0 | 27 0 | 10 5 (NOTE 11) | 1 70 | 2 50 | 1 0 | 24 0 | 26 0 | 7 0 | 6 5 | SEE NOTES 29 31, & 33 |
| CB-17 | RIPPLE (BYPASS) | 22 0 | 5 7 | 150 | 930 | 10K MINIMUM | 1 | SHORT, OPEN, OR 1 1 OHM | SOL | | 1 40 | 1 85 | 24 0 | 27 0 | 10 5 (NOTE 11) | 1 40 | 1 85 | 1 0 | 24 0 | 26 0 | 6 0 | 5 5 | SEE NOTES 29, 30, & 33 |
| CB-18 | RIPPLE (BYPASS) | 22 0 | 5 7 | 150 | 930 | 10K MINIMUM | 1 | SHORT, OPEN, OR 2 5 OHM | SOL | | 1 35 | 1 85 | 24 0 | 27 0 | 10 5 (NOTE 11) | 1 40 | 1 85 | 1 0 | 24 0 | 26 0 | 6 0 | 5 5 | SEE NOTES 29, 31 & 33 |

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TABLE II. Circuit board electrical test parameters. - Continued

Notes:

1. Abbreviations: MS - milliseconds; AMP - amperes; MA - milliamperes; QTY - quantity; MICROH - microhenry.
2. Tolerance: Resistance ± 0.1 ohm; voltage ± 1 percent; time ± 2 percent; current ± 2 percent; inductance ± 20 percent.
3. Temperature range: $+23^{\circ} \pm 5^{\circ}$C, unless otherwise specified.
4. Input voltage rise time: 0.1 milliseconds or less (except Tests no. CB-6 and CB-7).
5. Remove 1.1 or 2.5 ohm load after 12 ± 1.5 milliseconds, final load resistance shall be greater than 100K ohms.
6. Input line resistance includes line resistor, wire resistance, switch contact resistance, power source resistance, and test equipment resistance.
7. Fire and step load resistances includes load resistor, wire resistance, switch contact resistance, and test equipment resistance.
8. Use stepping switch Part Number M85242/2-1 when a solenoid is required as step load.
9. Input line inductance includes line inductor, wire inductance, power source inductance, and test equipment inductance.
10. Maximum fire output dwell is measured only when output load is not removed and shall not exceed 16.0 milliseconds. Minimum dwell time for a continuous load is 14.0 milliseconds.
11. Fire current "dwell time" (T O) does not apply to unloaded stations.
12. Even fire output "time to fire" measured from start of second input pulse.
13. Steady firing output shall occur within the time to fire listed with no indications of contact bounce on the firing output.
14. Fire current (I F) leakage on the fire output lines shall not exceed 50 milliamperes.
15. Solenoid current (I ST) leakage on the solenoid step lines shall not exceed 100 milliamperes.
16. Start reset pulse is the pulse that occurs for about 1/2 millisecond at the start of the input pulse and is measured at the reset test point.
17. Reset pulse delay (T RST) is a measure of the time from end of the input pulse to start of the reset pulse and is measured at the reset test point. Reset pulse delay shall range between 6.5 milliseconds (minimum) and 13.5 milliseconds (maximum).
18. Clock frequency (F OSC) is a measure of the output frequency of the internal clock at the clock test point. Clock frequency must fall within the range between 1560 hertz (minimum) and 1620 hertz (maximum).
19. The regulator voltage is a measure of the output voltage of the internal voltage regulator. The regulator voltage is measured at the voltage regulator test point. The regulator voltage must not exceed 9.0 volts (maximum).
20. The logic voltage is a measure of the voltage of the logic circuitry. The logic voltage is measured at the logic voltage test point. The logic voltage must not exceed 8.5 volts (maximum).
21. Preset voltage (V PR1) occurs coincident with the initial pulse and remains for 6.0 to 25 milliseconds. The preset voltage does not occur again until a new test is run (except Test CB-13).
22. Do not reset circuit board or test set after Tests CB-1A, 1B, or 1C. The four tests of CB-1 are run consecutively.
23. Typical switch noise (chatter) is shown in Figure 7.
24. Typical slow rise input is shown in Figure 7. For this test, rise time is 5.0 ± 0.2 MS.



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TABLE II. Circuit board electrical test parameters. - Continued

25. A 4 millisecond dropout can effect fire output "time to fire", fire "dwell time", step output "time to step" and step "dwell time".
26. A 4 millisecond dropout shall not cause the intervalometer to reset itself (advance to the next position) as would a new pulse.
27. Typical dropout is shown in Figure 8. For this test, dropout time is 4.0 ± 0.1 MS.
28. Dropout shall occur at 7 different times (minimum), with a maximum of one dropout per pulse.
29. A shorted fire output shall be 0.1 ohms and an unloaded output shall be greater than 100K ohms.
30. Short the odd output for the duration of two odd output pulses and short the even output for the duration of two even output pulses. Unload the odd output for the duration of two odd output pulses and unload the even output for the duration of two even output pulses. Load the odd and the even outputs for the remaining output pulses with 1.1 ohm loads.
31. Short the odd output for the duration of two odd output pulses and short the even output for the duration of two even output pulses. Unload the odd output for the duration of two odd output pulses and unload the even output for the duration of two even output pulses. Load the odd and the even outputs for the remaining output pulses with 2.5 ohm loads.
32. Preset voltage (V PR1) occurs coincident with the start of pulses 1, 2, 4, 6, 7, 9, 11, 12, 14, 16, 17, 19, and 21 and remains for 7.5 to 10 milliseconds at the beginning of these pulses (Test CB-13).
33. Ripple interval (Tests CB-14 through CB-18) shall range between 39.5 milliseconds (minimum) and 42.5 milliseconds (maximum).
34. The steady state input current (measured after the pulse has been applied for a minimum of 50 milliseconds, single mode, Tests CB-1A through CB-1D) shall be 0.20 ± 0.10 amperes.



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## TABLE III. DTU-29/A Circuit board troubleshooting guide.

THIS TABLE IS NOT A MANDATORY PART OF THE SPECIFICATION THE INFORMATION CONTAINED HEREIN IS INTENDED FOR GUIDANCE ONLY THIS TABLE COVERS SINGLE COMPONENT FAILURES ONLY AND ASSUMES CIRCUIT BOARD RUNS ARE NOT SHORTED TO OTHER RUNS OR COMPONENTS AND ARE CONTINUOUS ON EACH LAYER AND BETWEEN LAYERS A SINGLE COMPONENT FAILURE CAN HAVE MULTIPLE SYMPTOMS THE TEST PARAMETERS APPLY TO ROOM TEMPERATURE (23 +/- 5 DEG C)

| TEST NUMBERS | SYMPTOM/INDICATION | POSSIBLE CAUSES | CORRECTIVE ACTION |
|---|---|---|---|
| | GROUND CIRCUIT RESISTANCE | | |
| GROUND RESISTANCE TEST | RESISTANCE MEASURED BETWEEN WIRE FW1 AND/OR FW2 AND WIRE GN4 EXCEEDS SPEC | K2 FAILED | REPLACE K2 |
| | SHORTED INPUT | | |
| ALL TESTS | CIRCUIT BOARD FIRE INPUT CURRENT EXCEEDS 2 6 AMPERES WHEN FIRE PULSE IS APPLIED | CR30 OUT OF SPEC<br>C2 SHORTED<br>R1, 3, 5, 10, 11, 15, 16, 20, OR 51 OUT OF SPEC<br>Q1, 2, 3, OR 4 SHORTED | REPLACE CR30<br>REPLACE C2<br>REPLACE R1, 3, 5, 10, 11, 15, 16, 20, OR 51<br>REPLACE Q1, 2, 3, OR 4 |
| | REGULATOR VOLTAGE FAILURES | | |
| ALL TESTS | REGULATOR VOLTAGE EXCEEDS 9 0 VOLTS | CR20, 24, OR 25 OUT OF SPEC<br>R10 OUT OF SPEC<br>Q4 SHORTED | REPLACE CR20, 24, OR 25<br>REPLACE R10<br>REPLACE Q4 |
| ALL TESTS | REGULATOR VOLTAGE IS LESS THAN 6 0 VOLTS | CR10, 20, 24, OR 25 OUT OF SPEC<br>C3, 6, OR 17 SHORTED<br>R10 OR 22 OUT OF SPEC<br>Q4 FAILED<br>U1 OR 4 FAILED | REPLACE CR10, 20, 24, OR 25<br>REPLACE C3, 6, OR 17<br>REPLACE R10 OR 22<br>REPLACE Q4<br>REPLACE U1 OR 4 |
| | LOGIC VOLTAGE FAILURES | | |
| ALL TESTS | LOGIC VOLTAGE EXCEEDS 8 5 VOLTS | REGULATOR VOLTAGE IS TOO HIGH<br>CR31 SHORTED | CORRECT REGULATOR FAILURE<br>REPLACE CR31 |
| ALL TESTS | LOGIC VOLTAGE IS LESS THAN 5 5 VOLTS | REGULATOR VOLTAGE IS TOO LOW<br>CR31 OPEN<br>C13 OR 18 SHORTED<br>R19, 21, 29, OR 34 OUT OF SPEC<br>U2, 6, 7, 8, OR 9 FAILED | CORRECT REGULATOR FAILURE<br>REPLACE CR31<br>REPLACE C13 OR 18<br>REPLACE R19, 21, 29, OR 34<br>REPLACE U2, 6, 7, 8, OR 9 |
| | STEADY STATE INPUT CURRENT FAILURES | | |
| CB-1A THRU CB-1D | STEADY STATE INPUT CURRENT EXCEEDS 0 3 AMPERES | REGULATOR VOLTAGE IS TOO HIGH<br>LOGIC VOLTAGE IS TOO LOW<br>EXCESSIVE FIRE CURRENT LEAKAGE<br>EXCESSIVE STEP CURRENT LEAKAGE<br>CR10 OR 30 OUT OF SPEC<br>C2, 3, 6, OR 17 LEAKING<br>R1, 3, 5, 7, 10, 11, 15, 16, 17, 20, 22, OR 51 OUT OF SPEC<br>U1 OR 4 FAILED | CORRECT REGULATOR FAILURE<br>CORRECT LOGIC VOLTAGE FAILURE<br>CORRECT EXCESSIVE LEAKAGE<br>CORRECT EXCESSIVE LEAKAGE<br>REPLACE CR10 OR 30<br>REPLACE C2, 3, 6, OR 17<br>REPLACE R1, 3, 5, 7, 10, 11, 15, 16, 17, 20, 22, OR 51<br>REPLACE U1 OR 4 |
| CB-1A THRU CB-1D | STEADY STATE INPUT CURRENT IS LESS THAN 0 1 AMPERES | REGULATOR VOLTAGE IS TOO LOW<br>R35 OUT OF SPEC<br>K2 FAILED | CORRECT REGULATOR FAILURE<br>REPLACE R35<br>REPLACE K2 |



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TABLE III. DTU-29/A Circuit board troubleshooting guide. - Continued

| TEST NUMBERS | SYMPTOM/INDICATION | POSSIBLE CAUSES | CORRECTIVE ACTION |
|---|---|---|---|
| | OSCILLATOR FAILURES | | |
| ALL TESTS | OSCILLATOR DOESN'T OSCILLATE<br>OSC TEST POINT REMAINS HIGH FOR<br>DURATION OF FIRE PULSE (RESET - OK) | R38 OR 40 OUT OF SPEC<br>C4 SHORTED<br>U6, 8, OR 9 FAILED | REPLACE R38 OR 40<br>REPLACE C4<br>REPLACE U6, 8, OR 9 |
| ALL TESTS | OSCILLATOR DOESN'T OSCILLATE<br>OSC TEST POINT REMAINS LOW FOR<br>DURATION OF FIRE PULSE (RESET - OK) | R27, 38, 40, OR 41 OUT OF SPEC<br>C22 SHORTED<br>U6, 8, OR 9 FAILED | REPLACE R27, 38, 40, OR 41<br>REPLACE C22<br>REPLACE U6, 8, OR 9 |
| ALL TESTS | OSCILLATOR FREQUENCY IS TOO HIGH | | |
| | FREQUENCY IS BETWEEN 1620 AND 1660 HZ | C4 OR R40 OUT OF SPEC | REPLACE R40 WITH RLR07C1473FR |
| | FREQUENCY IS BETWEEN 1660 AND 1700 HZ | C4 OR R40 OUT OF SPEC | REPLACE R40 WITH RLR07C1503FR |
| | FREQUENCY IS GREATER THAN 1700 HZ | C4 OUT OF SPEC<br>R38, 40, OR 41 OUT OF SPEC<br>U6 FAILED | REPLACE C4<br>REPLACE R38, 40, OR 41<br>REPLACE U6 |
| ALL TESTS | OSCILLATOR FREQUENCY IS TOO LOW | | |
| | FREQUENCY IS BETWEEN 1540 AND 1560 HZ | C4 OR R40 OUT OF SPEC | REPLACE R40 WITH RLR07C1403FR |
| | FREQUENCY IS BETWEEN 1500 AND 1540 HZ | C4 OR R40 OUT OF SPEC | REPLACE R40 WITH RLR07C1373FR |
| | FREQUENCY IS LESS THAN 1500 HZ | C4 OUT OF SPEC<br>R38, 40, OR 41 OUT OF SPEC<br>U6 FAILED | REPLACE C4<br>REPLACE R38, 40, OR 41<br>REPLACE U6 |
| | RESET FAILURES | | |
| ALL TESTS | NO START RESET PULSE<br>(RESET PULSE - OK) | CR18 OUT OF SPEC OR OPEN<br>R6 OUT OF SPEC | REPLACE CR18<br>REPLACE R6 |
| ALL TESTS | RESET TEST POINT REMAINS LOW<br>(OSCILLATOR FREQUENCY - OK) | U6, 7, OR 8 FAILED | REPLACE U6, 7, OR 8 |
| ALL TESTS | RESET TEST POINT REMAINS HIGH<br>(OSCILLATOR FREQUENCY - OK) | R8 OR 29 OUT OF SPEC<br>C16 SHORTED<br>U6, 7, OR 8 FAILED | REPLACE R8 OR 29<br>REPLACE C16<br>REPLACE U6, 7, OR 8 |
| ALL TESTS | RESET TEST POINT REMAINS HIGH<br>OSCILLATOR TEST POINT REMAINS LOW | CR23 OUT OF SPEC OR OPEN<br>Q6 OUT OF SPEC OR OPEN<br>R13, 20, OR 25 OUT OF SPEC<br>C10 SHORTED<br>U1 OR 6 FAILED | REPLACE CR23<br>REPLACE Q6<br>REPLACE R13, 20, OR 25<br>REPLACE C10<br>REPLACE U1 OR 6 |
| ALL TESTS | RESET PULSE OCCURS TOO LATE (LONG) AFTER END OF FIRE PULSE (START RESET PULSE - OK) | | |
| | RESET PULSE IS 13.5 TO 16 MILLISECOND | C16 OR R8 OUT OF SPEC | REPLACE R8 WITH RLR07C6203GR |
| | RESET PULSE IS LONGER THAN 16 MILLI-SECONDS OR DOESN'T OCCUR | C10 OR 16 OUT OF SPEC<br>R8, 13, 20, 25, OR 29 OUT OF SPEC<br>U1, 6, 7, OR 8 OUT OF SPEC | REPLACE C10 OR 16<br>REPLACE R8, 13, 20, 25, OR 29<br>REPLACE U1, 6, 7, OR 8 |





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TABLE III. DTU-29/A Circuit board troubleshooting guide. - Continued

| TEST NUMBERS | SYMPTOM/INDICATION | POSSIBLE CAUSES | CORRECTIVE ACTION |
|---|---|---|---|
| RESET FAILURES - CONTINUED | | | |
| ALL TESTS | RESET PULSE OCCURS TOO SOON (SHORT) AFTER END OF FIRE PULSE (START RESET PULSE - OK) | | |
| | RESET PULSE IS 5 5 TO 6 5 MILLISECOND | C16 OR R8 OUT OF SPEC | REPLACE R8 WITH RLR07C9103GR |
| | RESET PULSE IS SHORTER THAN 5 5 MILLISECONDS | CR18 OUT OF SPEC<br>C16 OUT OF SPEC<br>R8 OUT OF SPEC<br>U6, 7, OR 8 OUT OF SPEC | REPLACE CR18<br>REPLACE C16<br>REPLACE R8<br>REPLACE U6, 7, OR 8 |
| LOGIC FAILURES | | | |
| ALL TESTS | NO STEP, FIRE 1 (ODD), OR FIRE 2 (EVEN) OUTPUTS (START RESET, RESET, OSCILLATOR, REGULATOR, AND LOGIC VOLTAGE - OK) | R33, 36, OR 49 OUT OF SPEC<br>C19 OR 21 OUT OF SPEC<br>U2, 6, 7, 8, OR 9 FAILED | REPLACE R33, 36, OR 49<br>REPLACE C19 OR 21<br>REPLACE U2, 6, 7, 8, OR 9 |
| ALL TESTS | STEP, FIRE 1 (ODD), FIRE 2 (EVEN) OUTPUTS, AND RIPPLE INTERVAL (TEST CB-14 THRU CB-18) ARE CONSISTANTLY SHORT (START RESET, RESET, OSCILLATOR, REGULATOR, AND LOGIC VOLTAGE - OK) | U2, 6, 7, OR 9 FAILED | REPLACE U2, 6, 7, OR 9 |
| ALL TESTS | STEP, FIRE 1 (ODD), FIRE 2 (EVEN) OUTPUTS, AND RIPPLE INTERVAL (TEST CB-14 THRU CB-18) ARE CONSISTANTLY LONG (START RESET, RESET, OSCILLATOR, REGULATOR, AND LOGIC VOLTAGE - OK) | U2, 6, 7, OR 9 FAILED | REPLACE U2, 6, 7, OR 9 |
| ALL TESTS | FIRE 1 (ODD) OUTPUT VOLTAGE REMAINS HIGH WHENEVER OUTPUT IS UNLOADED FOR BOTH ODD AND EVEN PULSES (STEP, FIRE 2 (EVEN), START RESET, RESET, OSCILLATOR, REGULATOR, AND LOGIC VOLTAGE - OK) | CR7 FAILED<br>Q1 FAILED<br>R5 OR 51 OUT OF SPEC | REPLACE CR7<br>REPLACE Q1<br>REPLACE R5 OR 51 |
| ALL TESTS | FIRE 2 (EVEN) OUTPUT VOLTAGE REMAINS HIGH WHENEVER OUTPUT IS UNLOADED FOR BOTH ODD AND EVEN PULSES (STEP, FIRE 1 (ODD), START RESET, RESET, OSCILLATOR, REGULATOR, AND LOGIC VOLTAGE - OK) | CR17 FAILED<br>Q2 FAILED<br>R15 OR 16 OUT OF SPEC | REPLACE CR17<br>REPLACE Q2<br>REPLACE R15 OR 16 |





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TABLE III. DTU-29/A Circuit board troubleshooting guide. - Continued

| TEST NUMBERS | SYMPTOM/INDICATION | POSSIBLE CAUSES | CORRECTIVE ACTION |
|---|---|---|---|
| STEP AND FIRE FAILURES | | | |
| ALL TESTS | FIRE 1 (ODD) AND FIRE 2 (EVEN) ARE IN REVERSE ORDER, STEP PRECEDES FIRE 1 | PRESET SIGNAL FAILURE<br>CR26 FAILED<br>R32, 37, OR 46 OUT OF SPEC<br>C7 OR 8 SHORTED<br>U8 FAILED | CHECK PRESET WIRE (PR1)<br>REPLACE CR26<br>REPLACE R32, 37, OR 46<br>REPLACE C7 OR 8<br>REPLACE U8 |
| ALL TESTS | FIRE 1 (ODD) FIRES FOR DURATION OF EACH ODD PULSE, FIRE 2 (EVEN) FIRES FOR DURATION OF EACH EVEN PULSE (STEP - OK) | Q5 OR Q10 FAILED<br>CR22, 27, OR 29 FAILED<br>R3, 24, 39, 44, OR 45 OUT OF SPEC<br>U1 OR 6 FAILED | REPLACE Q5 OR Q10<br>REPLACE CR22, 27, OR 29<br>REPLACE R3, 24, 39, 44, OR 45<br>REPLACE U1 OR 6 |
| ALL TESTS | OUTPUTS STAY IN STEP AND FIRE 1 (ODD) FOR ALL PULSES | Q8 OR 9 FAILED<br>CR3 FAILED<br>R16, 28, 30, 43, 47, OR 50 OUT OF SPEC<br>K1 FAILED<br>U1, 2, 4, 6, 7, OR 8 FAILED | REPLACE Q8 OR 9<br>REPLACE CR3<br>REPLACE R16, 28, 30, 43, 47, OR 50<br>REPLACE K1<br>REPLACE U1, 2, 4, 6, 7, OR 8 |
| ALL TESTS | OUTPUTS STAY IN FIRE 2 (EVEN) WITH NO STEP AFTER THE FIRST PULSE | Q8 OR 9 FAILED<br>CR13 FAILED<br>R31, 43, 47, OR 50 OUT OF SPEC<br>C9 FAILED<br>K1 FAILED<br>U1 2, 4, 6, 7, OR 8 FAILED | REPLACE Q8 OR 9<br>REPLACE CR13<br>REPLACE R31, 43, 47, OR 50<br>REPLACE C9<br>REPLACE K1<br>REPLACE U1, 2, 4, 6, 7, OR 8 |
| ALL TESTS | NO FIRE 1 (ODD) OR FIRE 2 (EVEN) (STEP - OK) | Q5 SHORTED OR Q10 OPEN<br>CR10 FAILED<br>R3, 24, 35, 39, 44, 45, OR 49 OUT OF SPEC<br>C5, 19, OR 23 SHORTED<br>K2 FAILED<br>U1 OR 6 FAILED | REPLACE Q5 OR Q10<br>REPLACE CR10<br>REPLACE R3, 24, 35, 39, 44, 45 OR 49<br>REPLACE C5, 19, OR 23<br>REPLACE K2<br>REPLACE U1 OR 6 |
| CB-1A, 1C, AND CB-2 THRU CB-1B | NO STEP (FIRE 1 (ODD) AND FIRE 2 (EVEN) - OK) | CR21 OUT OF SPEC<br>R9, 22, 23, OR 26 OUT OF SPEC<br>Q3 OR 7 OUT OF SPEC<br>C14 OUT OF SPEC<br>U1 OR 4 FAILED | REPLACE CR21<br>REPLACE R9, 22, 23, OR 26<br>REPLACE Q3 OR 7<br>REPLACE C14<br>REPLACE U1 OR 4 |
| ALL TESTS (MOST NOTICABLE WHEN INPUT VOLTAGE EXCEEDS 27 VOLTS) | 1 MILLISECOND CURRENT SPIKE APPEARS ON FIRE 1 (ODD) OR FIRE 2 (EVEN) FIRING LINE AT ABOUT 5 MS PRIOR TO START OF FIRING PULSE ON SAME LINE | Q5 OUT OF SPEC | REPLACE Q5 |
| ALL TESTS | CURRENT SPIKE APPEARS ON FIRE 1 (ODD) AND FIRE 2 (EVEN) FIRING LINE AND/OR ON STEP LINE AT START OF INPUT PULSES | CR9, 19, 27, OR 28 OUT OF SPEC<br>R7, 9, 17, OR 49 OUT OF SPEC<br>C5, 14, OR 19 OUT OF SPEC<br>Q5 OUT OF SPEC<br>K2 FAILED | REPLACE CR9, 19, 27, OR 28<br>REPLACE R7, 9, 17, OR 49<br>REPLACE C5, 14, OR 19<br>REPLACE Q5<br>REPLACE K2 |





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TABLE III. DTU-29/A Circuit board troubleshooting guide. - Continued

| TEST NUMBERS | SYMPTOM/INDICATION | POSSIBLE CAUSES | CORRECTIVE ACTION |
|---|---|---|---|
| FIRE 1 (ODD) FAILURES | | | |
| ALL TESTS | FIRE 1 (ODD) FIRING CURRENT TOO HIGH | CR4 OUT OF SPEC<br>R1, 2, OR 5 OUT OF SPEC<br>Q1 OUT OF SPEC | REPLACE CR4<br>REPLACE R1, 2, OR 5<br>REPLACE Q1 |
| ALL TESTS | FIRE 1 (ODD) FIRING CURRENT TOO LOW | CR4, 6, OR 7 OUT OF SPEC<br>C5 OR C19 OUT OF SPEC<br>Q1, 5, OR 10 OUT OF SPEC<br>R1, 2, 3, 4, 39, 44, OR 45 OUT OF SPEC<br>K1 OR 2 FAILED<br>U1 FAILED | REPLACE CR4, 6, OR 7<br>REPLACE C5 OR 19<br>REPLACE Q1, 5, OR 10<br>REPLACE R1, 2, 3, 4, 39, 44, 45<br>REPLACE K1 OR 2<br>REPLACE U1 |
| ALL TESTS | FIRE 1 (ODD) FIRES WITH EACH PULSE (FIRE 2 (EVEN) AND STEP - OK) | CR7 OR 8 FAILED<br>R1, 5, OR 51 OUT OF SPEC<br>K1 FAILED<br>Q1 FAILED | REPLACE CR7 OR 8<br>REPLACE R1, 5, OR 51<br>REPLACE K1<br>REPLACE Q1 |
| ALL TESTS | FIRE 1 (ODD) FIRES FOR DURATION OF EACH ODD PULSE (FIRE 2 (EVEN) AND STEP - OK) | CR5 FAILED<br>Q1 FAILED | REPLACE CR5<br>REPLACE Q1 |
| ALL TESTS | LEAKAGE CURRENT EXCEEDS 50 MILLI-AMPERES | CR5, 22, OR 29 OUT OF SPEC<br>R1, 3, 5, OR 39 OUT OF SPEC<br>U1 FAILED<br>Q1, 5, OR 10 OUT OF SPEC | REPLACE CR5, 22, OR 29<br>REPLACE R1, 3, 5, OR 39<br>REPLACE U1<br>REPLACE Q1, 5, OR 10 |
| FIRE 2 (EVEN) FAILURES | | | |
| ALL TESTS | FIRE 2 (EVEN) FIRING CURRENT TOO HIGH | CR14 OUT OF SPEC<br>R11, 12, OR 15 OUT OF SPEC<br>Q2 OUT OF SPEC | REPLACE CR14<br>REPLACE R11, 12, OR 15<br>REPLACE Q2 |
| ALL TESTS | FIRE 2 (EVEN) FIRING CURRENT TOO LOW | CR14, 16, OR 17 OUT OF SPEC<br>C5 OR C19 OUT OF SPEC<br>Q2, 5, OR 10 OUT OF SPEC<br>R3, 11, 12, 14, 39, 44, OR 45 OUT OF SPEC<br>K1 OR 2 FAILED<br>U1 FAILED | REPLACE CR14, 16, OR 17<br>REPLACE C5 OR 19<br>REPLACE Q2, 5, OR 10<br>REPLACE R3, 11, 12, 14, 39, 44, OR 45<br>REPLACE K1 OR 2<br>REPLACE U1 |
| ALL TESTS | FIRE 2 (EVEN) FIRES WITH EACH PULSE (FIRE 1 (ODD) AND STEP - OK) | CR15 OR 17 FAILED<br>R11, 15, OR 16 OUT OF SPEC<br>Q2 FAILED<br>K1 FAILED | REPLACE CR15 OR 17<br>REPLACE R11, 15, OR 16<br>REPLACE Q2<br>REPLACE K1 |
| ALL TESTS | FIRE 2 (EVEN) FIRES FOR DURATION OF EACH EVEN PULSE (FIRE 1 (ODD) AND STEP - OK) | CR11 FAILED<br>Q2 FAILED | REPLACE CR11<br>REPLACE Q2 |
| ALL TESTS | LEAKAGE CURRENT EXCEEDS 50 MILLI-AMPERES | CR11, 22, OR 29 OUT OF SPEC<br>R3, 11, 15, OR 39 OUT OF SPEC<br>U1 FAILED<br>Q2, 5, OR 10 OUT OF SPEC | REPLACE CR11, 22, OR 29<br>REPLACE R3, 11, 15, OR 39<br>REPLACE U1<br>REPLACE Q2, 5, OR 10 |





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TABLE III. DTU-29/A Circuit board troubleshooting guide. - Continued

| TEST NUMBERS | SYMPTOM/INDICATION | POSSIBLE CAUSES | CORRECTIVE ACTION |
|---|---|---|---|
| SOLENOID STEP FAILURES | | | |
| ALL TESTS | SOLENOID STEP CURRENT TOO HIGH | REGULATOR VOLTAGE IS TOO HIGH<br>CR21 OUT OF SPEC<br>R22 OR 23 OUT OF SPEC<br>Q3 SHORTED | CORRECT REGULATOR FAILURE<br>REPLACE CR21<br>REPLACE R22 OR 23<br>REPLACE Q3 |
| ALL TESTS | SOLENOID STEP CURRENT TOO LOW | REGULATOR VOLTAGE IS TOO LOW<br>CR21 OUT OF SPEC<br>C14 OR 19 OUT OF SPEC<br>R9, 22, OR 23 OUT OF SPEC<br>Q3 OR 7 OUT OF SPEC<br>U1 FAILED | CORRECT REGULATOR FAILURE<br>REPLACE CR21<br>REPLACE C14 OR 19<br>REPLACE R9, 22, OR 23<br>REPLACE Q3 OR 7<br>REPLACE U1 |
| ALL TESTS | SOLENOID STEPS WITH EACH PULSE (FIRE 1 (ODD) AND FIRE 2 (EVEN) - OK) | R22 OR 26 OUT OF SPEC<br>Q3 OR 7 FAILED<br>U1 OR U4 FAILED | REPLACE R22 OR 26<br>REPLACE Q3 OR 7<br>REPLACE U1 OR U4 |
| ALL TESTS | SOLENOID STEPS FOR DURATION OF EACH ODD PULSE (FIRE 1 (ODD) AND FIRE 2 (EVEN) - OK) | U4 FAILED | REPLACE U4 |
| ALL TESTS | LEAKAGE CURRENT EXCEEDS 100 MILLI-AMPERES | REGULATOR VOLTAGE IS TOO HIGH<br>R22 OR 26 OUT OF SPEC<br>Q3 OR 7 OUT OF SPEC<br>U1 FAILED | CORRECT REGULATOR FAILURE<br>REPLACE R22 OR 26<br>REPLACE Q3 OR 7<br>REPLACE U1 |
| SINGLE/RIFFLE FAILURES | | | |
| CR-1A THRU CR-1D | RIFFLE INOPERATIVE (OUTPUT LIMITED TO ONE STEP, ONE FIRE 1 (ODD) PULSE, AND NO FIRE 2 (EVEN) PULSES (RIFFLE MODE SELECTED) | C15 OR 20 SHORTED<br>R18 OR 34 OUT OF SPEC<br>U8 FAILED | REPLACE C15 OR 20<br>REPLACE R18 OR 34<br>REPLACE U8 |
| CR-1A THRU CR-1D | RIFFLE OPERATES CONTINUOUSLY (MULTIPLE STEP AND FIRE PULSES OCCUR WITH EACH INPUT PULSE - SINGLE MODE SELECTED) | SINGLE/RIFFLE SELECT FAILURE<br>R18 OR 34 OUT OF SPEC<br>U8 FAILED | CHECK RIFFLE SELECT WIRE (SS1)<br>REPLACE R18 OR 34<br>REPLACE U8 |
| SELF-GENERATED EMI/NOISE/OSCILLATION | | | |
| ALL TESTS | OSCILLATION ON INPUT POWER LINE, FIRE OUTPUT LINES, AND REGULATOR VOLTAGE TEST POINT FREQUENCY ABOUT 1.4 MHZ | C23 FAILED<br>R35 FAILED | REPLACE C23<br>REPLACE R35 |





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1. Rocket launcher, 2.75 inch, 19 tube (LAU-61A/A).
2. Rocket launcher, 2.75 inch, 7 tube (LAU-68B/A).
3. All aircraft that carry LAU-61A/A and LAU-68B/A rocket launchers.

Preparing Activity:
Navy - AS

(Project 5930-N569-1)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-I-85581/1(AS) | INTERVALOMETER, ROCKET LAUNCHER, REUSABLE, NINETEEN (19) OUTPUT, DIRECT LINE OPERATED |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
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| | ☐ USER |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)* _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
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